All orders

Gyropup retained prototype workstreams · Revision: cd276824fcbe0bea6c3c39099120afd465b751a3

All parts: 279 of 279

Original leg, rig, turret and earlier body planning lists. Current front/rear-pack robot manufacturing plan remains incomplete. Checklist marks track parts, not purchased quantities. This list does not qualify fit, strength or a complete robot.

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/ballast_shaft	Tapped linear shaft 1 in x 6 in, 1050 steel, 3/8-16 x 3/4 in deep both ends ballast_shaft	6649K433 seller's no. 6649K433 McMaster-Carr	2	$52.94	$105.88	1 alternate · 1 drop-in DROP-IN McMaster-Carr 4162N31 + 6435K18 collar · $38.42 checked 2026-10-27	Uses and sourcing notes rig ×2 the two ballast posts	☑	☐
☐	buy/circlip_50	External Retaining Ring, for 50 mm OD, Black-Phosphate Spring Steel circlip_50	98541A167 McMaster-Carr	1 buy 1 pack of 5	$3.10 $15.50 per 5	$15.50 used $3.10	See note below	Uses and sourcing notes turret ×1 circlip_50	☐	☑
Alternates: 5 alternates · 5 drop-in DROP-IN Metric Specialties 50NCRE · $0.71 · available (Add to cart enabled; no count shown) checked 2026-10-27 DROP-IN Aspen Fasteners ME304-50-00050 · Now: $183.21 (Box of 50) · Contact Support for Availability Info checked 2026-10-27 DROP-IN Fastenal 1135344 (package) / 35344 (bulk) checked 2026-10-27 DROP-IN McMaster-Carr 90154A768 · $3.37 per pack of 1 · Delivers tomorrow 7-9 am Zinc-chromate plated, sold singly: the same DIN 471-50 ring and groove. checked 2026-10-27 DROP-IN McMaster-Carr 90967A350 · $5.59 per pack of 1 · Delivers tomorrow 7-9 am 1.4122 stainless, sold singly: the same DIN 471-50 ring and groove. checked 2026-10-27										
☐	buy/circlip_60	External Retaining Ring, for 60 mm OD, Black-Phosphate Spring Steel circlip_60	98541A174 McMaster-Carr Delivers Wednesday 7-9 am (2026-10-27)	1	$3.05	$3.05	See note below	Uses and sourcing notes leg ×1 holds the apex 61812 on the wheel link's hub	☐	☐
Alternates: 4 alternates · 4 drop-in DROP-IN Aspen Fasteners ME304-60-00040 · Now: $177.63 (Box of 40) · Contact Support for Availability Info checked 2026-10-27 DROP-IN Fastenal 35351 checked 2026-10-27 DROP-IN McMaster-Carr 90154A811 · $4.41 per pack of 1 · Delivers tomorrow 7-9 am Zinc-chromate plated instead of black phosphate: the same DIN 471-60 ring and groove. checked 2026-10-27 DROP-IN McMaster-Carr 90967A375 · $7.90 per pack of 1 · Delivers tomorrow 7-9 am 1.4122 stainless instead of spring steel: the same DIN 471-60 ring and groove; does not rust. checked 2026-10-27										

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/dowel_5x16	Dowel Pin, 52100 Alloy Steel, 5 mm Diameter, 16 mm Long dowel_5x16	91595A352 McMaster-Carr	6 buy 1 pack of 50	$0.3844 $19.22 per 50	$19.22 used $2.31	See note below	Uses and sourcing notes leg ×4 into each motor's two locating holes, 12 mm into the hip frame turret ×2 hw_tm_dowels	☐	☐
Alternates: 3 alternates · 3 drop-in DROP-IN WDS Components 662-20316 · $0.60 · Stock Guaranteed; stock availability < 500 checked 2026-10-27 DROP-IN Aspen Fasteners ME397-5X16-00300 · Now: $180.47 (Box of 300) · Contact Support for Availability Info checked 2026-10-27 DROP-IN Aspen Fasteners UB006-5X16-000400 · Now: $437.89 (Box of 400) · Contact Support for Availability Info checked 2026-10-27										
☐	buy/dowel_5x20	Dowel Pin, 52100 Alloy Steel, 5 mm Diameter, 20 mm Long dowel_5x20	91595A356 McMaster-Carr	4 buy 1 pack of 25	$0.4352 $10.88 per 25	$10.88 used $1.74	NO ALTERNATE	Uses and sourcing notes leg ×4 two in each adapter's standard face, pressed 8 mm into its H7 holes, 12 mm into the upper link (issue #5)	☐	☐
☐	buy/eye_sleeve	Metric Press-Fit Drill Bushing, 12 mm ID, 18 mm OD, 20 mm Long, Chamfer on One End eye_sleeve	96990A403 McMaster-Carr Delivers tomorrow 7-9 am (2026-10-27)	4	$16.11	$64.44	See note below	Uses and sourcing notes leg ×4 steel sleeves, two end to end in each upper link's knee eye: spread the pin's load on the plastic	☐	☐
Alternates: 3 alternates · 1 drop-in DROP-IN JW Winco DIN 179-B12-20-A-NI checked 2026-10-27 CHANGES Reid Supply DM-12025 · $7.88 (1+) · In stock (9) 25 mm long instead of 20 mm (same 12 mm ID and 18 mm OD): two end to end make 50 mm, so the knee eye must be 10 mm deeper, or one 25 mm with one 15 mm. ID and OD tolerance classes (F7 / n6) are not stated. checked 2026-10-27 CHANGES McMaster-Carr 96990A240 · $37.07 each · Delivers tomorrow 7-9 am 12 x 16 x 10 DIN 179 bushing: four end to end fill the knee eye's 40 mm, in a Ø16 bore instead of Ø18, so the upper link's press bore changes from Ø17.96 to Ø15.96 (cad/fivebar.py EYE_SLEEVE). Four per eye at $37.07 each. checked 2026-10-27										

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/grommet_cable_pass	Buna-N Rubber Grommet, for 11/16" Hole Diameter and 1/4" Material Thickness, 7/16" ID grommet_cable_pass	9307K884 McMaster-Carr	1 buy 1 pack of 10	$1.08 $10.82 per 10	$10.82 used $1.08	1 alternate · 1 drop-in DROP-IN printed product page · $0.00 · print a split TPU 95A grommet (as the deck's lidar pass) checked 2026-10-27	Uses and sourcing notes p1 frame hardware ×1 the turret harness through the deck	☐	☐
☐	buy/insert_m3	Heat-Set Threaded Inserts for Plastic, Standard, Brass, M3 x 0.5 mm Thread Size, 5.7 mm Installed Length insert_m3	94459A140 McMaster-Carr	97 buy 2 packs of 50	$0.2182 $10.91 per 50	$21.82 used $21.17	See note below	See note below	☐	☐

Alternates: 6 alternates · 6 drop-in DROP-IN ruthex RX-M3x5.7 (SKU GE-M3x57-001) · InStock (schema.org availability) checked 2026-10-27 DROP-IN 3DJake CNC-X001FCE4M5 · $11.18 ($0.11 / pc), excl. VAT · In stock checked 2026-10-27 DROP-IN 3DJake CNC-X001FCE4M5 · $11.18 / 100 · In stock checked 2026-10-27 DROP-IN ruthex RX-M3x5.7 (SKU GE-M3x57-001) · InStock (schema.org availability) checked 2026-10-27 DROP-IN 3DJake CNC-X001FCE4M5 · $11.18 / 100 · In stock checked 2026-10-27 DROP-IN ruthex RX-M3x5.7 (SKU GE-M3x57-001) · InStock (schema.org availability) checked 2026-10-27

Part details: Uses and sourcing notes turret ×41 hw_insert_m3 stand ×8 bench p1 frame hardware ×26 P1: 4 power dcdc, 2 bus insulator 1, 2 bus bar 1, 2 bus insulator 2, 2 bus bar 2, 5 bay side l high, 5 bay side r high, 1 bay door front l, 1 bay door front r, 1 bay door rear l, 1 bay door rear r (cad/body_joints.py) p3 shell ×22 P3: 2 wall l front, 2 wall l rear, 2 wall r front, 2 wall r rear, 2 shell front post l, 2 shell front post r, 2 d435 cradle front feet, 2 shell rear post l, 2 shell rear post r, 2 d435 cradle rear feet, 2 eyes clip (cad/body_joints.py)

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/insert_m5	Heat-Set Threaded Inserts for Plastic, Standard, Brass, M5 x 0.8 mm Thread Size, 7.9 mm Installed Length insert_m5	94459A775 seller's no. 94459A775 McMaster-Carr	49 buy 2 packs of 25	$0.2728 $6.82 per 25	$13.64 used $13.37	See note below	See note below	☐	☐
Alternates: 8 alternates · 6 drop-in CHANGES ruthex RX-M5x9.5 (SKU GE-M5x95-001) · InStock (schema.org availability) NOT drop-in: 9.5 mm long vs McMaster 7.9 mm (1.6 mm longer). The short version RX-M5Sx5.8 (and CNC Kitchen 'M5 x 5.8 short', also listed) is 2.1 mm shorter. No 7.9 mm M5 at ruthex or CNC Kitchen. EUR 9.99 per 50 (VAT incl.), no USD price. checked 2026-10-27 CHANGES 3DJake CNC-B09FXM8ZMV · $11.18 ($0.22 / pc), excl. VAT · In stock 5.8 mm long instead of 7.9 mm, so 2.1 mm less thread engagement and pull-out strength. Same 6.4 mm hole per CNC Kitchen's table. It fits the existing 6.4 mm hole without a CAD change and sits flush with spare depth below. checked 2026-10-27 DROP-IN ruthex RX-M5x9.5 (SKU GE-M5x95-001) · InStock (schema.org availability) checked 2026-10-27 DROP-IN 3DJake CNC-B09FXM8ZMV · $11.18 / 50 · In stock checked 2026-10-27 DROP-IN 3DJake CNC-B09FXM8ZMV · $11.18 / 50 · In stock checked 2026-10-27 DROP-IN ruthex RX-M5x9.5 (SKU GE-M5x95-001) · InStock (schema.org availability) checked 2026-10-27 DROP-IN 3DJake CNC-B09FXM8ZMV · $11.18 / 50 · In stock checked 2026-10-27 DROP-IN ruthex RX-M5x9.5 (SKU GE-M5x95-001) · InStock (schema.org availability) checked 2026-10-27 **Part details:** Uses and sourcing notes leg ×6 the frame's flange ends (the body, or the rig's carriage) bench ×3 B4: one in each M5 puller stand ×4 bench p1 frame hardware ×32 P1: 6 hip frame left, 6 hip frame right, 1 post 1 foot, 1 post 1 head, 1 post 2 foot, 1 post 2 head, 1 post 3 foot, 1 post 3 head, 1 post 4 foot, 1 post 4 head, 4 riser p1, 1 plate post 1 foot, 1 plate post 1 head, 1 plate post 2 foot, 1 plate post 2 head, 1 plate post 3 foot, 1 plate post 3 head, 1 plate post 4 foot, 1 plate post 4 head (cad/body_joints.py) p2 head ×4 P2: 4 riser (cad/body_joints.py)									
☐	buy/knee_pin	Alloy Steel Shoulder Screws, 12 mm Shoulder Diameter, 90 mm Shoulder Length, M10 x 1.5 mm Thread knee_pin	92981A411 McMaster-Carr Delivers tomorrow 7-9 am (2026-10-27)	2	$8.02	$16.04	See note below	Uses and sourcing notes leg ×2 knee shoulder screws, 12 mm x 90, M10 (cad/fivebar.py knee_pin_length)	☐	☐
Alternates: 1 alternate · 1 drop-in DROP-IN Aspen Fasteners ME343-1015X90X12-00010 · $168.80 (box of 10) · Contact Support for Availability Info ISO 7379 12 x 90, M10 x 1.5, class 12.9 but plain (no black oxide): it rusts sooner. Sold only by the box of 10. checked 2026-10-27										
☐	buy/knee_spacer	Multipurpose 6061 Aluminum Round Tube, 0.065" Wall Thickness, 5/8" OD knee_spacer	9056K68 McMaster-Carr	2 buy 1 length, cut 4 per length	$6.26 / length $1.57 a piece	$6.26 used $3.13	See note below	Uses and sourcing notes leg ×2 between each knee's two inner races (24 mm)	☐	☐
Alternates: 3 alternates · 3 drop-in DROP-IN Speedy Metals t61r.625x.065 · 12in length: $8.96 (custom by the inch: Base Price $0.50) checked 2026-10-27 DROP-IN Alcobra Metals 22210129bf34 · $20.03 (12in) · In stock checked 2026-10-27 DROP-IN McMaster-Carr 90018A130 · $70.39 each · Delivers in 1-3 weeks Made to order, no lathe needed: order it 0.630 in OD x 0.500 in ID x 1.102 in (28.0 mm) long (16.0 x 12.7 mm, under the 17 mm limit, clears the 12 mm shoulder). About $70 each at 1-4 against $1.56 of tube, and 1-3 weeks. checked 2026-10-27										

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/nut_m10_nyloc	High-Strength Steel Nylon-Insert Locknut, Zinc Plated, M10 x 1.5 mm Thread Size nut_m10_nyloc	94645A220 McMaster-Carr Delivers Wednesday 7-9 am (2026-10-27)	2 buy 1 pack of 25	$0.4632 $11.58 per 25	$11.58 used $0.93	2 alternates · 2 drop-in DROP-IN Belmetric NN10CLZ · $0.26 · In stock checked 2026-10-27 DROP-IN Aspen Fasteners ME022-1015 · Now: $148.51 · Contact Support for Availability Info checked 2026-10-27	Uses and sourcing notes leg ×2 knee pins, in the upper links' pockets	☐	☐
☐	buy/nut_m3	Steel Hex Nut, Medium-Strength, Class 8, M3 x 0.5 mm (held in the P3 shell's nut pads) nut_m3	90592A085 seller's no. 90592A085 McMaster-Carr product page read (McMaster ships next day) (2026-10-27)	10 buy 1 pack of 100	$0.0294 $2.94 per 100	$2.94 used $0.29	1 alternate · 1 drop-in DROP-IN McMaster-Carr 91828A211 · $5.68 / 100 · 18-8 stainless M3 hex nut, the same size checked 2026-10-27	Uses and sourcing notes p3 shell ×10 P3: 2 splice l, 2 splice r, 4 shell door, 2 shell hatch (cad/body_joints.py)	☐	☐
☐	buy/nut_m5_nyloc	High-Strength Steel Nylon-Insert Locknut, Zinc Plated, M5 x 0.8 mm Thread Size nut_m5_nyloc	94645A102 McMaster-Carr	10 buy 1 pack of 100	$0.1566 $15.66 per 100	$15.66 used $1.57	2 alternates · 2 drop-in DROP-IN Belmetric NN5CLZ · $0.09 · in stock checked 2026-10-27 DROP-IN Belmetric NN5CLZ · $0.09 · in stock checked 2026-10-27	Uses and sourcing notes turret ×2 hw_hub_nuts p3 shell ×8 P3: 4 ear bracket l, 4 ear bracket r (cad/body_joints.py)	☐	☐
☐	buy/nut_m6_nyloc	High-Strength Steel Nylon-Insert Locknut, Zinc Plated, M6 x 1 mm Thread Size nut_m6_nyloc	94645A205 McMaster-Carr	10 buy 1 pack of 100	$0.1226 $12.26 per 100	$12.26 used $1.23	1 alternate · 1 drop-in DROP-IN Global Industrial WBB2298457 (Titan Fasteners AAV06) · $25.95 / 100 · ships in 2 business days: DIN 985 M6 in 304 stainless (the same size, 6 mm tall) checked 2026-10-27	Uses and sourcing notes p1 frame hardware ×10 held in the bus insulators' hex pockets under the bars' holes (they cannot turn)	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/pin_lanyard	Lanyard, 3/64 in cable, loop-to-eye, 12 in pin_lanyard	90312A296 seller's no. 90312A296 McMaster-Carr	1	$3.26	$3.26	1 alternate · 1 drop-in DROP-IN Essentra Components LN3/64120003BGP · $2.90 · 900 In stock checked 2026-10-27	Uses and sourcing notes rig ×1 keeps the pin on the drop bracket	☐	☐
☐	buy/quick_pin	Ball-lock quick-release pin with D-ring, 3/8 in, 1-1/2 in usable, 17-4 PH quick_pin	92384A274 seller's no. 92384A274 McMaster-Carr	1	$38.21	$38.21	1 alternate · 1 drop-in DROP-IN McMaster-Carr 92384A760 (1-3/4 in usable) · $40.73 checked 2026-10-27	Uses and sourcing notes rig ×1 the drop test's release (S4)	☐	☐
☐	buy/screw_3_8x1	Alloy steel socket head screw 3/8-16 x 1 in screw_3_8x1	91251A624 seller's no. 91251A624 McMaster-Carr	2 buy 1 pack of 25	$0.4992 $12.48 per 25	$12.48 used $1.00	1 alternate · 1 drop-in DROP-IN McMaster-Carr 91251A622 (3/4 in, no washer) · $11.17 / 25 checked 2026-10-27	Uses and sourcing notes rig ×2 each post through the plate (17.8 mm into the 19 mm thread with the washer)	☐	☐
☐	buy/screw_3_8x3_4	Alloy steel socket head screw 3/8-16 x 3/4 in screw_3_8x3_4	91251A622 seller's no. 91251A622 McMaster-Carr	2 buy 1 pack of 25	$0.4468 $11.17 per 25	$11.17 used $0.89	1 alternate · 1 drop-in DROP-IN Bolt Depot 5044 · $0.34 · no count shown; orderable (Add button, not marked 'Temporarily unavailable') checked 2026-10-27	Uses and sourcing notes rig ×2 the posts' free ends, with the 2 in washers that hold the plates on	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/screw_m3x16	Black-Oxide Alloy Steel Socket Head Screw, M3 x 0.5 x 16 mm (the P3 shell: walls and faces into the posts' inserts, the door, hatch and splices) screw_m3x16	91290A120 seller's no. 91290A120 McMaster-Carr product page read (McMaster ships next day) (2026-10-27)	16 buy 1 pack of 100	$0.1222 $12.22 per 100	$12.22 used $1.96	1 alternate · 1 drop-in DROP-IN Belmetric BSH3X16 · $0.16 · in stock: DIN 912 12.9 black, the same screw; $0.10 each from 25 checked 2026-10-27	Uses and sourcing notes p3 shell ×16 P3: 2 wall l front, 2 wall l rear, 2 wall r front, 2 wall r rear, 2 shell front post l, 2 shell front post r, 2 shell rear post l, 2 shell rear post r (cad/body_joints.py)	☐	☐
☐	buy/screw_m3x6	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M3 x 0.5 mm Thread Size, 6 mm Long screw_m3x6	91290A111 McMaster-Carr	45 buy 1 pack of 100	$0.1159 $11.59 per 100	$11.59 used $5.22	3 alternates · 3 drop-in DROP-IN Belmetric BSH3X6 · $0.12 · in stock checked 2026-10-27 DROP-IN Belmetric BSH3X6 · $0.12 · in stock checked 2026-10-27 DROP-IN Belmetric BSH3X6 · $0.12 · in stock checked 2026-10-27	See note below	☐	☐
Part details: Uses and sourcing notes turret ×20 hw_screw_m3x6 p1 frame hardware ×20 P1: 4 power dcdc, 1 bay side l low, 5 bay side l high, 1 bay side r low, 5 bay side r high (cad/body_joints.py); 4 the J501 carrier onto its standoffs p3 shell ×5 P3: 2 eyes clip, 3 fan duct (cad/body_joints.py)										

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/screw_m3x8_socket	Black-oxide alloy steel socket head screw, M3 x 0.5 x 8 mm (the bus bars to their insulators, the insulators to the plate) screw_m3x8_socket	91290A113 seller's no. 91290A113 McMaster-Carr product page read (McMaster ships next day) (2026-10-27)	26 buy 1 pack of 100	$0.1328 $13.28 per 100	$13.28 used $3.45	2 alternates · 2 drop-in DROP-IN McMaster-Carr 91290A115 · $13.23 / 100 · M3 x 10 of the same series checked 2026-10-27 DROP-IN McMaster-Carr 91290A115 · $13.23 / 100 · M3 x 10 of the same series checked 2026-10-27	See note below	☐	☐
Part details: Uses and sourcing notes p1 frame hardware ×8 P1: 2 bus insulator 1, 2 bus bar 1, 2 bus insulator 2, 2 bus bar 2 (cad/body_joints.py) p3 shell ×18 P3: 2 splice l, 2 splice r, 2 d435 cradle front feet, 2 d435 front back, 2 d435 cradle rear feet, 2 d435 rear back, 4 shell door, 2 shell hatch (cad/body_joints.py)										
☐	buy/screw_m4x10_flat	Black-oxide alloy steel flat head hex drive screw, 90 deg, M4 x 0.7 x 10 mm (the spine standoffs, flush on the frame side) screw_m4x10_flat	91294A190 seller's no. 91294A190 McMaster-Carr product page read (McMaster ships next day) (2026-10-27)	16 buy 1 pack of 100	$0.0638 $6.38 per 100	$6.38 used $1.02	1 alternate · 1 drop-in DROP-IN McMaster-Carr 91294A188 · $6.25 / 100 · M4 x 8 of the same series (engages 3.3 mm in the standoff) checked 2026-10-27	Uses and sourcing notes p1 frame hardware ×16 P1: 8 standoff left, 8 standoff right (cad/body_joints.py)	☐	☐
☐	buy/screw_m4x12_socket	Black-oxide alloy steel socket head screw, M4 x 0.7 x 12 mm screw_m4x12_socket	91290A148 seller's no. 91290A148 McMaster-Carr product page read (McMaster ships next day) (2026-10-27)	14 buy 1 pack of 100	$0.123 $12.30 per 100	$12.30 used $1.72	See note below	Uses and sourcing notes turret ×4 hw_screw_m4x12 p1 frame hardware ×10 P1: 2 dcdc bracket, 4 adapter plate front, 4 adapter plate rear (cad/body_joints.py)	☐	☐
Alternates: 2 alternates · 2 drop-in DROP-IN McMaster-Carr 91290A150 · $14.23 / 100 · M4 x 14 of the same series (a stack of three lugs) checked 2026-10-27 DROP-IN McMaster-Carr 91290A150 · $14.23 / 100 · M4 x 14 of the same series (a stack of three lugs) checked 2026-10-27										

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/screw_m4x16_rail	18-8 stainless socket head screw M4 x 16 screw_m4x16_rail	91292A118 seller's no. 91292A118 McMaster-Carr	36 buy 1 pack of 100	$0.1163 $11.63 per 100	$11.63 used $4.19	1 alternate · 1 drop-in DROP-IN Bolt Depot 23082 (class 12.9, zinc) · $0.19 checked 2026-10-27	Uses and sourcing notes rig ×36 the rails into the drop-in nuts (HIWIN's M4 x 16)	☐	☐
☐	buy/screw_m4x20	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M4 x 0.7 mm Thread Size, 20 mm Long screw_m4x20	91290A168 McMaster-Carr	32 buy 1 pack of 100	$0.1447 $14.47 per 100	$14.47 used $4.63	2 alternates · 2 drop-in DROP-IN Belmetric BSH4X20 · $0.20 · In stock checked 2026-10-27 DROP-IN Belmetric BSH4X20 · $0.20 · In stock checked 2026-10-27	See note below	☐	☐
Part details: Uses and sourcing notes leg ×20 motor lugs (10 per motor), seats 8.5 mm down: 4.4 mm nominal penetration into 5 mm-min usable threads; measure before assembly p1 frame hardware ×12 P1: 1 isolator 1, 1 isolator 2, 1 isolator 3, 1 isolator 4, 4 bay rail l, 4 bay rail r (cad/body_joints.py)										
☐	buy/screw_m5x10_button	Black-oxide class 10.9 button head hex drive screw, ISO 7380, M5 x 0.8 x 10 mm (tray, belly, posts, VESCs, bay) screw_m5x10_button	91239A224 seller's no. 91239A224 McMaster-Carr product page read (McMaster ships next day) (2026-10-27)	18 buy 1 pack of 100	$0.1179 $11.79 per 100	$11.79 used $2.12	2 alternates · 2 drop-in DROP-IN McMaster-Carr 91239A228 · $12.06 / 100 · M5 x 12 of the same series checked 2026-10-27 DROP-IN McMaster-Carr 91239A228 · $12.06 / 100 · M5 x 12 of the same series checked 2026-10-27	See note below	☐	☐
Part details: Uses and sourcing notes p1 frame hardware ×16 P1: 1 plate post 1 foot, 1 plate post 1 head, 1 plate post 2 foot, 1 plate post 2 head, 1 plate post 3 foot, 1 plate post 3 head, 1 plate post 4 foot, 1 plate post 4 head, 4 plate vesc r flange, 4 plate vesc l flange (cad/body_joints.py) p2 head ×2 P2: 2 turret plate ends (cad/body_joints.py)										

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/screw_m5x12_button	Black-oxide class 10.9 button head hex drive screw, ISO 7380, M5 x 0.8 x 12 mm (the hip frames into the cheeks) screw_m5x12_button	91239A228 seller's no. 91239A228 McMaster-Carr product page read (McMaster ships next day) (2026-10-27)	40 buy 1 pack of 100	$0.1206 $12.06 per 100	$12.06 used $4.82	1 alternate · 1 drop-in DROP-IN McMaster-Carr 91239A224 · $11.79 / 100 · M5 x 10 of the same series (engages 5.3 mm: only if the 12 is out) checked 2026-10-27	See note below	☐	☐
Part details: Uses and sourcing notes p1 frame hardware ×40 P1: 6 hip frame left, 4 tray flange left, 4 belly flange left, 6 hip frame right, 4 tray flange right, 4 belly flange right, 1 post 1 foot, 1 post 1 head, 1 post 2 foot, 1 post 2 head, 1 post 3 foot, 1 post 3 head, 1 post 4 foot, 1 post 4 head, 4 riser p1 (cad/body_joints.py)										
☐	buy/screw_m5x14_socket	Black-oxide alloy steel socket head screw, M5 x 0.8 x 14 mm (the fuse board's plate bond, MH1, into the plate's flush nut) screw_m5x14_socket	91290A230 seller's no. 91290A230 McMaster-Carr product page read (McMaster ships next day) (2026-10-27)	8 buy 1 pack of 100	$0.1743 $17.43 per 100	$17.43 used $1.39	1 alternate · 1 drop-in DROP-IN McMaster-Carr 91290A228 · $16.36 / 100 · M5 x 12 of the same series, with MH1's standoff printed 1.6 mm shorter checked 2026-10-27	Uses and sourcing notes p3 shell ×8 P3: 4 ear bracket l, 4 ear bracket r (cad/body_joints.py)	☐	☐
☐	buy/screw_m5x16	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M5 x 0.8 mm Thread Size, 16 mm Long screw_m5x16	91290A232 seller's no. 91290A232 McMaster-Carr	10 buy 1 pack of 100	$0.1727 $17.27 per 100	$17.27 used $1.73	2 alternates · 2 drop-in DROP-IN Belmetric BSH5X16 · $0.17 · in stock checked 2026-10-27 DROP-IN Bolt Depot 6522 · $0.14 · no count shown; orderable (Add button, not marked 'Temporarily unavailable') checked 2026-10-27	Uses and sourcing notes rig ×6 the plate onto the hip frame's six M5 inserts (with washers) turret ×4 hw_tm_screws	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/screw_m5x25	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M5 x 0.8 mm Thread Size, 25 mm Long screw_m5x25	91290A252 seller's no. 91290A252 McMaster-Carr	36 buy 1 pack of 50	$0.2192 $10.96 per 50	$10.96 used $7.89	1 alternate · 1 drop-in DROP-IN Bolt Depot 6524 (M5 x 25 class 12.9 black oxide) · $0.15 checked 2026-10-27	Uses and sourcing notes leg ×18 motor outputs through the CNC output adapters (9 per motor), heads in pockets below the face bench ×18 the dyno: each hub onto its motor's output (9 each)	☐	☐
☐	buy/screw_m5x40	Black-Oxide Alloy Steel Socket Head Screw, M5 x 0.8 mm Thread Size, 40 mm Long, Partially Threaded screw_m5x40	91290A662 seller's no. 91290A662 McMaster-Carr	19 buy 1 pack of 50	$0.1618 $8.09 per 50	$8.09 used $3.07	See note below	Uses and sourcing notes leg ×16 upper links onto the output adapters (8 each), heads proud bench ×3 B4: the screws under test (two inserts to 3.5 kN, the link-under-the-head test)	☐	☐
Alternates: 4 alternates · 4 drop-in DROP-IN Belmetric BSH5X40 · $0.28 · In stock checked 2026-10-27 DROP-IN McMaster-Carr 91290A258 · $16.58 per pack of 50 · Delivers tomorrow 7-9 am The US-origin column of the same class 12.9 ISO 4762 table: same screw, about twice the price. checked 2026-10-27 DROP-IN Belmetric BSH5X40 · $0.28 · In stock checked 2026-10-27 DROP-IN McMaster-Carr 91290A258 · $16.58 / 50 · Delivers tomorrow 7-9 am checked 2026-10-27										
☐	buy/screw_m6x16	Black-Oxide Alloy Steel Socket Head Screw, M6 x 1 mm Thread Size, 16 mm Long, Fully Threaded screw_m6x16	91290A667 McMaster-Carr	10 buy 1 pack of 100	$0.0859 $8.59 per 100	$8.59 used $0.86	1 alternate · 1 drop-in DROP-IN Belmetric BSH6X16 · $0.20 · in stock: DIN 912 class 12.9, black oxide, fully threaded (the same screw); $0.12 each from 25 checked 2026-10-27	Uses and sourcing notes p1 frame hardware ×10 the bus bars' lug stacks: down through the lug(s) and the bar into the insulator's nut	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/screw_m6x30	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M6 x 1 mm Thread Size, 30 mm Long screw_m6x30	91290A332 seller's no. 91290A332 McMaster-Carr	12 buy 1 pack of 50	$0.2272 $11.36 per 50	$11.36 used $2.73	2 alternates · 2 drop-in DROP-IN Belmetric BSH6X30 · $0.23 · In stock checked 2026-10-27 DROP-IN Belmetric BSH6X30 · $0.23 · In stock checked 2026-10-27	See note below	☐	☐
Part details: Uses and sourcing notes leg ×4 the BotWheel's flange, from inside the wheel link's hub (M6 x 40 bottoms out unless its threads are > 15 deep: measure) rig ×8 the deck into the members' top slots (18 mm ply + washer + the slot's lip, 7.6 mm into the nut)										
☐	buy/screw_m6x35	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M6 x 1 mm Thread Size, 35 mm Long screw_m6x35	91290A334 seller's no. 91290A334 McMaster-Carr	4 buy 1 pack of 50	$0.2422 $12.11 per 50	$12.11 used $0.97	1 alternate · 1 drop-in DROP-IN Bolt Depot 6543 · $0.24 · no count shown; orderable (Add button, not marked 'Temporarily unavailable') checked 2026-10-27	Uses and sourcing notes rig ×4 the bottom bumpers' bases, counterbored 14 mm	☐	☐
☐	buy/shoulder_6x12_m5	Alloy Steel Shoulder Screws, 6 mm Shoulder Diameter, 12 mm Shoulder Length, M5 x 0.8 mm Thread shoulder_6x12_m5	92981A101 McMaster-Carr	1	$2.82	$2.82	See note below	Uses and sourcing notes turret ×1 shoulder_screw	☐	☐
Alternates: 2 alternates · 1 drop-in DROP-IN Aspen Fasteners ME343-508X12X6-00080 · Now: $191.06 (Box of 80) · Contact Support for Availability Info checked 2026-10-27 CHANGES Belmetric BSHLDR5X6X10/19 · $4.39 · In stock Shoulder length is 10 mm instead of 12 mm (2 mm shorter). The clamped stack must be 2 mm thinner or the CAD changed. The 6 mm shoulder, M5 x 0.8 thread and class 12.9 black oxide all match. checked 2026-10-27										
☐	buy/standoff_m3x16_ff	Lightweight aluminium female-threaded hex standoff, 6 mm hex, M3 x 16 mm (the J501 carrier on the pallet) standoff_m3x16_ff	95947A048 seller's no. 95947A048 McMaster-Carr product page read (McMaster ships next day) (2026-10-27)	4	$0.810	$3.24	1 alternate · 1 drop-in DROP-IN McMaster-Carr 95947A051 · $0.81 · M3 x 19 of the same series (the carrier 3 mm higher) checked 2026-10-27	Uses and sourcing notes p1 frame hardware ×4 the J501 carrier on the compute pallet	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/standoff_m4x31_ff	Lightweight aluminium female-threaded hex standoff, 6 mm hex, M4 x 31 mm (the spine: between the cheeks) standoff_m4x31_ff	95947A065 seller's no. 95947A065 McMaster-Carr product page read (McMaster ships next day) (2026-10-27)	8	$1.46	$11.68	1 alternate · 1 drop-in DROP-IN McMaster-Carr 95947A062 · $1.35 · M4 x 30 of the same series, with a printed 1 mm washer at one end checked 2026-10-27	Uses and sourcing notes p1 frame hardware ×8 the spine: between the cheeks in the hip frames' bolt rows	☐	☐
☐	buy/stop_pin	Vented Pull-Out Dowel Pin with Flat Shaft, High-Strength 52100 Alloy Steel, 16 mm Diameter, 40 mm Long stop_pin	98432A386 McMaster-Carr Delivers Thursday (2026-10-27)	4	$5.88	$23.52	See note below	Uses and sourcing notes leg ×4 stop posts: Ø16 x 40 vented pull-out dowels, 23 mm into blind holes in the hip frame (epoxy-bonded, D74), TPU sleeves over them	☐	☐
Alternates: 2 alternates · none drop-in CHANGES McMaster-Carr 91595A845 · $5.01 each The plain 52100 dowel (DIN 6325-m6, Rockwell C52, Ø16 x 40): no vent and no tapped end. Print a Ø2 vent through the hole's 1.1 mm floor at its outer edge (off the motor's face) so air and compound get out, and accept that the pin cannot be pulled out of the blind hole. checked 2026-10-27 CHANGES Aspen Fasteners ME397-16X40-00008 · $192.49 (box of 8) · Contact Support for Availability Info A plain DIN 6325 dowel (m6, hardened Rockwell C58-C62, rounded ends): the same changes as the plain McMaster pin; four times the price. checked 2026-10-27										
☐	buy/tool-circlip-pliers	External retaining-ring pliers for a 50 mm ring (tool) tool-circlip-pliers	5449A83 seller's no. 5449A83 McMaster-Carr 'Delivers tomorrow 7-9 am' (McMaster's in-stock wording) (2026-10-27)	1	$37.25	$37.25	1 alternate · 1 drop-in DROP-IN McMaster-Carr 5449A82 · $33.48 · listed in the category table (not opened; McMaster stocks the family) checked 2026-10-27	Uses and sourcing notes tools ×1 tool	☐	☐
☐	buy/tool-dial-indicator	Dial indicator with a jointed-arm magnetic base (tool) tool-dial-indicator	5518N11 seller's no. 5518N11 McMaster-Carr 'Delivers tomorrow 7-9 am' (2026-10-27)	1	$79.62	$79.62	1 alternate · 1 drop-in DROP-IN McMaster-Carr 6302N12 · $84.93 · 'Delivers tomorrow 7-9 am' checked 2026-10-27	Uses and sourcing notes tools ×1 tool	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/tool-insert-tips	Heat-set insert installation tip set, M2-M8, for a Hakko 900M-style iron (tool) tool-insert-tips	8915N111 seller's no. 8915N111 McMaster-Carr 'Delivers tomorrow 7-9 am' (2026-10-27)	1	$22.66	$22.66	See note below	Uses and sourcing notes tools ×1 tool	☐	☐
Alternates: 2 alternates · 2 drop-in DROP-IN McMaster-Carr 97338A140 · $74.43 · 'Delivers tomorrow 7-9 am' checked 2026-10-27 DROP-IN McMaster-Carr 92160A315 + 92160A323 + 92160A327 · $28.14 / 3 · listed in the category table (product pages not opened) checked 2026-10-27										
☐	buy/tool-reamer-5mm	Chucking reamer, 5 mm, turned by hand in a tap wrench (tool) tool-reamer-5mm	8851A18 seller's no. 8851A18 McMaster-Carr listed in the category table (product page not opened) (2026-10-27)	1	$26.23	$26.23	1 alternate · 1 drop-in DROP-IN McMaster-Carr 2822A37 · $106.53 · 'Delivers tomorrow 7-9 am' checked 2026-10-27	Uses and sourcing notes tools ×1 tool	☐	☐
☐	buy/turret_dowel_4x8	Dowel Pin, 52100 Alloy Steel, 4 mm Diameter, 8 mm Long turret_dowel_4x8	91595A150 McMaster-Carr	2 buy 1 pack of 50	$0.242 $12.10 per 50	$12.10 used $0.48	1 alternate · 1 drop-in DROP-IN WDS Components 662-20208 · $0.49 · stock availability < 500 checked 2026-10-27	Uses and sourcing notes turret ×2 hw_dowel_4x8	☐	☐
☐	buy/turret_insert_m3_short	Heat-Set Threaded Inserts for Plastic, Standard, Brass, M3 x 0.5 mm Thread Size, 4.3 mm Installed Length turret_insert_m3_short	94459A130 McMaster-Carr	2 buy 1 pack of 50	$0.2182 $10.91 per 50	$10.91 used $0.44	1 alternate · 1 drop-in DROP-IN ruthex RX-M3Sx4.0 (SKU GE-M3Sx40-002) · InStock (schema.org availability) checked 2026-10-27	Uses and sourcing notes stand ×2 bench	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/turret_insert_m4	Heat-Set Threaded Inserts for Plastic, Standard, Brass, M4 x 0.7 mm Thread Size, 6.4 mm Installed Length turret_insert_m4	94459A160 seller's no. 94459A160 McMaster-Carr	16 buy 1 pack of 50	$0.2346 $11.73 per 50	$11.73 used $3.75	See note below	Uses and sourcing notes rig ×7 the pod: the contactor (2), the relay (1), the pendant clip (2); the ToF target arm (2) turret ×9 hw_insert_m4	☐	☐
Alternates: 2 alternates · 1 drop-in CHANGES ruthex RX-M4x8.1 (SKU GE-M4x81-001) · InStock (schema.org availability) NOT drop-in: 8.1 mm long vs McMaster 6.4 mm (1.7 mm longer, needs a deeper boss). The other ruthex option, RX-M4Sx4.0 short, is 2.4 mm shorter (page not opened). Neither ruthex nor CNC Kitchen (M4x8.1 / M4x4) sells a 6.4 mm M4. EUR 9.49 per 50 (VAT incl.), no USD price. checked 2026-10-27 DROP-IN Fastenal 11125996 (E-Z LOK / E-Z Sonic TH-M40-SV) · $0.24 · Local DC Inventory; Direct Ship: Ships TOMORROW checked 2026-10-27										
☐	buy/turret_magnet_disc	Neodymium Magnet, Magnetized Through Thickness, 1/8" Thick, 3/16" OD turret_magnet_disc	5862K102 McMaster-Carr	1	$1.00	$1.00	1 alternate · 1 drop-in DROP-IN K&J Magnetics D32-N52 · $0.35 · In Stock checked 2026-10-27	Uses and sourcing notes turret ×1 index_magnet	☐	☐
☐	buy/turret_nut_m2_5	Corrosion-Resistant 18-8 Stainless Steel Hex Nut, M2.5 x 0.45 mm Thread Size turret_nut_m2_5	91828A113 McMaster-Carr	2 buy 1 pack of 100	$0.071 $7.10 per 100	$7.10 used $0.14	1 alternate · 1 drop-in DROP-IN Belmetric NR2.5SS · $0.06 · in stock checked 2026-10-27	Uses and sourcing notes stand ×2 bench	☐	☐
☐	buy/turret_oring_ct	Oil-Resistant Buna-N O-Ring, 1/16 Fractional Width, Dash Number 006 turret_oring_ct	9452K14 McMaster-Carr	1 buy 1 pack of 100	$0.0279 $2.79 per 100	$2.79 used $0.03	1 alternate · 1 drop-in DROP-IN The O-Ring Store LLC B70006 · $0.03 · 368780 in Stock checked 2026-10-27	Uses and sourcing notes turret ×1 hw_oring_ct	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/turret_screw_m2_5x8	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M2.5 x 0.45 mm Thread Size, 8 mm Long turret_screw_m2_5x8	91290A102 seller's no. 91290A102 McMaster-Carr	6 buy 1 pack of 50	$0.246 $12.30 per 50	$12.30 used $1.48	See note below	Uses and sourcing notes electronics ×4 the Feather's FeatherWing into the pod's bosses (self-tapping into PETG) stand ×2 bench	☐	☐
Alternates: 2 alternates · 2 drop-in DROP-IN The Nutty Company MSHC2.58 · $0.12 · in stock (no count shown) checked 2026-10-27 DROP-IN Bolt Depot 13330 · $0.20 · no count shown; orderable (Add button, not marked 'Temporarily unavailable') checked 2026-10-27										
☐	buy/turret_screw_m2x10	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M2 x 0.4 mm Thread Size, 10 mm Long turret_screw_m2x10	91290A017 McMaster-Carr	4 buy 1 pack of 100	$0.1589 $15.89 per 100	$15.89 used $0.64	1 alternate · 1 drop-in DROP-IN The Nutty Company MSHC210 · $0.13 · in stock (no count shown) checked 2026-10-27	Uses and sourcing notes turret ×4 hw_screw_m2x10	☐	☐
☐	buy/turret_screw_m2x20	Black-Oxide Alloy Steel Socket Head Screw, M2 x 0.4 mm Thread Size, 20 mm Long turret_screw_m2x20	91290A049 McMaster-Carr	4 buy 1 pack of 100	$0.1658 $16.58 per 100	$16.58 used $0.66	1 alternate · 1 drop-in DROP-IN The Nutty Company MSHC220 · $0.18 · in stock (no count shown) checked 2026-10-27	Uses and sourcing notes turret ×4 hw_screw_m2x20	☐	☐
☐	buy/turret_screw_m2x6_flat	18-8 Stainless Steel Hex-Drive Flat Head Screw, M2 x 0.4 mm Thread Size, 6 mm Long turret_screw_m2x6_flat	92125A052 McMaster-Carr	4 buy 1 pack of 50	$0.240 $12.00 per 50	$12.00 used $0.96	1 alternate · 1 drop-in DROP-IN Metric Specialties 2C6MXF2 · $0.07 · available checked 2026-10-27	Uses and sourcing notes turret ×4 hw_flat_m2x6	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/turret_screw_m3x10	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M3 x 0.5 mm Thread Size, 10 mm Long turret_screw_m3x10	91290A115 McMaster-Carr	10 buy 1 pack of 100	$0.1323 $13.23 per 100	$13.23 used $1.32	2 alternates · 2 drop-in DROP-IN Belmetric BSH3X10 · $0.14 · in stock checked 2026-10-27 DROP-IN Belmetric BSH3X10 · $0.14 · in stock checked 2026-10-27	Uses and sourcing notes turret ×2 hw_screw_m3x10 p1 frame hardware ×8 P1: 2 bay pad 1, 2 bay pad 2, 2 bay pad 3, 2 bay pad 4 (cad/body_joints.py)	☐	☐
☐	buy/turret_screw_m3x10_flat	Black-Oxide Alloy Steel Hex-Drive Flat Head Screw, 90 Degree Countersink, M3 x 0.5 mm Thread Size, 10 mm Long turret_screw_m3x10_flat	91294A130 McMaster-Carr	3 buy 1 pack of 100	$0.0636 $6.36 per 100	$6.36 used $0.19	1 alternate · 1 drop-in DROP-IN Belmetric SF3X10 · $0.18 · in stock checked 2026-10-27	Uses and sourcing notes turret ×3 hw_flat_m3x10	☐	☐
☐	buy/turret_screw_m3x14_flat_ss	18-8 Stainless Steel Hex-Drive Flat Head Screw, M3 x 0.5 mm Thread Size, 14 mm Long turret_screw_m3x14_flat_ss	92125A133 McMaster-Carr	4 buy 1 pack of 100	$0.1065 $10.65 per 100	$10.65 used $0.43	1 alternate · 1 drop-in DROP-IN Belmetric SF3X14CLZ · $0.18 · in stock checked 2026-10-27	Uses and sourcing notes turret ×4 hw_flat_m3x14	☐	☐
☐	buy/turret_screw_m3x16	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M3 x 0.5 mm Thread Size, 16 mm Long turret_screw_m3x16	91290A120 McMaster-Carr	4 buy 1 pack of 100	$0.1222 $12.22 per 100	$12.22 used $0.49	1 alternate · 1 drop-in DROP-IN Belmetric BSH3X16 · $0.16 · in stock checked 2026-10-27	Uses and sourcing notes turret ×4 hw_screw_m3x16	☐	☐
☐	buy/turret_screw_m3x6_flat	Black-Oxide Alloy Steel Hex-Drive Flat Head Screw, 90 Degree Countersink, M3 x 0.5 mm Thread Size, 6 mm Long turret_screw_m3x6_flat	91294A126 McMaster-Carr	6 buy 1 pack of 100	$0.0545 $5.45 per 100	$5.45 used $0.33	2 alternates · 2 drop-in DROP-IN Belmetric SF3X6 · $0.18 · in stock checked 2026-10-27 DROP-IN Belmetric SF3X6 · $0.18 · in stock checked 2026-10-27	Uses and sourcing notes turret ×2 hw_flat_m3x6 p1 frame hardware ×4 the carrier standoffs up through the pallet's countersinks	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/turret_screw_m3x8	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M3 x 0.5 mm Thread Size, 8 mm Long turret_screw_m3x8	91290A113 McMaster-Carr	23 buy 1 pack of 100	$0.1328 $13.28 per 100	$13.28 used $3.05	1 alternate · 1 drop-in DROP-IN Belmetric BSH3X8 · $0.13 · in stock checked 2026-10-27	Uses and sourcing notes turret ×19 hw_screw_m3x8 stand ×4 bench	☐	☐
☐	buy/turret_screw_m3x8_button	Class 10.9 Screw, M3 x 0.5 mm Thread Size, 8 mm Long turret_screw_m3x8_button	91239A113 McMaster-Carr	8 buy 1 pack of 100	$0.0982 $9.82 per 100	$9.82 used $0.79	1 alternate · 1 drop-in DROP-IN Fastener Superstore 227258 · $31.01 / carton of 100 · not shown checked 2026-10-27	Uses and sourcing notes turret ×8 hw_button_m3x8	☐	☐
☐	buy/turret_screw_m3x8_flat	Black-Oxide Alloy Steel Hex-Drive Flat Head Screw, 90 Degree Countersink, M3 x 0.5 mm Thread Size, 8 mm Long turret_screw_m3x8_flat	91294A128 McMaster-Carr	4 buy 1 pack of 100	$0.0582 $5.82 per 100	$5.82 used $0.23	1 alternate · 1 drop-in DROP-IN Belmetric SF3X8 · $0.18 · in stock checked 2026-10-27	Uses and sourcing notes turret ×4 hw_flat_m3x8	☐	☐
☐	buy/turret_screw_m4x10	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M4 x 0.7 mm Thread Size, 10 mm Long turret_screw_m4x10	91290A144 seller's no. 91290A144 McMaster-Carr	24 buy 1 pack of 100	$0.1355 $13.55 per 100	$13.55 used $3.25	2 alternates · 2 drop-in DROP-IN Belmetric BSH4X10 · $0.14 · in stock checked 2026-10-27 DROP-IN Bolt Depot 6506 · $0.12 · no count shown; orderable (Add button, not marked 'Temporarily unavailable') checked 2026-10-27	Uses and sourcing notes rig ×20 the four blocks onto the plate (16: 3.65 mm into the 5 mm threads, 0.9 d); the pendant clip (2) and the ToF arm (2) into inserts turret ×4 hw_screw_m4x10	☐	☐
☐	buy/turret_setscrew_m3x6_nylon	Nylon-Tip 18-8 Stainless Steel Set Screw, M3 x 0.5 mm Thread, 6 mm Long turret_setscrew_m3x6_nylon	93285A115 McMaster-Carr	4 buy 1 pack of 5	$3.46 $17.31 per 5	$17.31 used $13.85	1 alternate · 1 drop-in DROP-IN Belmetric SS3X6NYTPSS · $3.25 · in stock checked 2026-10-27	Uses and sourcing notes turret ×4 hw_setscrew_m3x6	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/turret_standoff_m3x12	Aluminum Male-Female Threaded Hex Standoff, 4.5mm Hex, 12mm Long, M3 x 0.50 mm Thread turret_standoff_m3x12	98952A110 McMaster-Carr	4	$0.830	$3.32	1 alternate · 1 drop-in DROP-IN Mouser 761-M2107-3005-AL · $0.53 · 691 checked 2026-10-27	Uses and sourcing notes turret ×4 hw_standoff_m3x12	☐	☐
☐	buy/turret_standoff_m3x14	Aluminum Male-Female Threaded Hex Standoff, 4.5mm Hex, 14mm Long, M3 x 0.50 mm Thread turret_standoff_m3x14	98952A112 McMaster-Carr	4	$0.840	$3.36	1 alternate · 1 drop-in DROP-IN Mouser 761-M2109-3005-AL · $1.42 · 73 checked 2026-10-27	Uses and sourcing notes turret ×4 hw_standoff_m3x14	☐	☐
☐	buy/turret_thumbscrew_m4x12	Steel Raised Knurled-Head Thumb Screw, M4 x 0.7mm Thread Size, 12mm Long turret_thumbscrew_m4x12	92581A250 McMaster-Carr	1	$4.64	$4.64	1 alternate · 1 drop-in DROP-IN Belmetric BTHUMB4X12SS · $2.46 · in stock checked 2026-10-27	Uses and sourcing notes turret ×1 ct_thumbscrew	☐	☐
☐	buy/turret_tube_coupling	Continuous-Flex Tpv Rubber Tube for Chemicals, 3/16" ID, 5/16" OD turret_tube_coupling	51225K24 McMaster-Carr	1	$0.620	$0.62	1 alternate · 1 drop-in DROP-IN Zoro Zoro Select ZUSA-HT-3487 (Zoro # G8076328) · $12.49 checked 2026-10-27	Uses and sourcing notes turret ×1 ring_coupling	☐	☐
☐	buy/washer_2in	Oversized washer 0.406 in ID x 2.000 in OD washer_2in	91117A219 seller's no. 91117A219 McMaster-Carr	2 buy 1 pack of 5	$2.72 $13.59 per 5	$13.59 used $5.44	1 alternate · 1 drop-in DROP-IN Fastenal 11103742 · $33.00 / 20 · Local DC Inventory; Direct Ship: Ships TOMORROW checked 2026-10-27	Uses and sourcing notes rig ×2 retain the weight plates	☐	☐
☐	buy/washer_3_8	Oversized washer for 3/8 in, 1.25 in OD washer_3_8	91090A121 seller's no. 91090A121 McMaster-Carr	2 buy 1 pack of 50	$0.1424 $7.12 per 50	$7.12 used $0.28	1 alternate · 1 drop-in DROP-IN Bolt Depot 2970 · $0.19 · no count shown; orderable (Add button, not marked 'Temporarily unavailable') checked 2026-10-27	Uses and sourcing notes rig ×2 under the post screws on the plate's front	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/washer_m12_thin	18-8 Stainless Steel Washer, Undersized, for M12 Screw Size, 13.0 mm ID, 20.0 mm OD washer_m12_thin	98689A118 McMaster-Carr Delivers Wednesday 7-9 am (2026-10-27)	4 buy 1 pack of 100	$0.1371 $13.71 per 100	$13.71 used $0.55	2 alternates · 2 drop-in DROP-IN Belmetric WF12X20SS · $0.20 · In stock checked 2026-10-27 DROP-IN Aspen Fasteners ME351-12 · Now: $179.53 · Contact Support for Availability Info checked 2026-10-27	Uses and sourcing notes leg ×4 each side of each knee's bearing stack	☐	☐
☐	buy/washer_m3_large	18-8 Stainless Steel Oversized Washer, for M3 Screw, 3.2mm ID, 9.0mm OD, 0.7mm to 0.9mm Thick washer_m3_large	91116A120 McMaster-Carr	11 buy 1 pack of 100	$0.0481 $4.81 per 100	$4.81 used $0.53	See note below	Uses and sourcing notes turret ×3 hw_washer_m3_large p1 frame hardware ×8 P1: 2 bay pad 1 (washer), 2 bay pad 2 (washer), 2 bay pad 3 (washer), 2 bay pad 4 (washer) (cad/body_joints.py)	☐	☐
Alternates: 4 alternates · 4 drop-in DROP-IN Belmetric WFE3X9SS · $0.08 · in stock checked 2026-10-27 DROP-IN Aspen Fasteners ME380-3 · Now: $180.21 · Contact Support for Availability Info checked 2026-10-27 DROP-IN Belmetric WFE3X9SS · $0.08 · in stock checked 2026-10-27 DROP-IN Aspen Fasteners ME380-3 · Contact Support for Availability Info checked 2026-10-27										

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/washer_m5	18-8 Stainless Steel General Purpose Washer, for M5 Screw Size, 5.3 mm ID, 10.0 mm OD washer_m5	93475A240 seller's no. 93475A240 McMaster-Carr	6 buy 1 pack of 100	$0.0424 $4.24 per 100	$4.24 used $0.25	2 alternates · 2 drop-in DROP-IN Belmetric WF5SS · $0.07 · in stock checked 2026-10-27 DROP-IN Bolt Depot 4515 · $0.05 · no count shown; orderable (Add button, not marked 'Temporarily unavailable') checked 2026-10-27	Uses and sourcing notes rig ×6 under the plate screws	☐	☐
☐	buy/washer_m6_din433	Washer, undersized (DIN 433), for M6 screw, 6.4 mm ID, 11.0 mm OD, 1.6 mm thick, 18-8 stainless steel washer_m6_din433	product page NO PART NUMBER McMaster-Carr	4	NO PRICE	—	NO ALTERNATE	Uses and sourcing notes leg ×4 under the BotWheel screws' heads on the wheel link's foot (D74: spreads the clamp; 1.8 N m)	☐	☐
☐	buy/washer_m6_large	18-8 Stainless Steel Oversized Washer, for M6 Screw, 6.4mm ID, 18.0mm OD, 1.4mm to 1.8mm Thick washer_m6_large	91116A150 seller's no. 91116A150 McMaster-Carr	10 buy 1 pack of 50	$0.294 $14.70 per 50	$14.70 used $2.94	See note below	Uses and sourcing notes rig ×8 under the deck screws bench ×2 B4: under the screw under test, over the cage's hole	☐	☐
Alternates: 2 alternates · 2 drop-in DROP-IN Bolt Depot 7315 · $0.12 · no count shown; orderable (Add button, not marked 'Temporarily unavailable') checked 2026-10-27 DROP-IN Bolt Depot 7315 · $0.12 · no count shown; orderable (Add button, not marked 'Temporarily unavailable') checked 2026-10-27										

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/robstride-rs-04	RobStride RS-04 robstride-rs-04	RS-04 Seeed Studio ships from CN	2	$255.00	$510.00	See note below	See note below	☐	☐

Alternates: 8 alternates · 4 drop-in DROP-IN OpenELAB ROBSTRIDE04 · $297.99 (variant 'Motor (w/o Debug)'; Debug Board $24.99 extra) · Orderable; 8-15 business days; ships from China; duties included checked 2026-10-27 DROP-IN AIFIT Lab ROBSTRIDE 04 Motor - Only Motor · $255.00 (Only Motor); $273.00 with USB-to-CAN module · Backordered checked 2026-10-27 DROP-IN Robotics Center Robstride RS-04 (no SKU shown) · $295.00 · Backordered; 4-6 weeks checked 2026-10-27 DROP-IN rcdrone.top RobStride-04-1 · $319.00 (120nm Motor); $329.00 with CAN2USB debugger · Orderable checked 2026-10-27 CHANGES OpenELAB DM-J10010L-2EC · $321.99 · Orderable; ships from China Closest substitute (Damiao DM-J10010L-2EC): same 40 N·m rated / 120 N·m peak and 200 rpm no-load at 48 V, but rated output speed is 100 rpm at 48 V vs the RS-04's 167 rpm. Round body ~120 mm dia x 53 mm (RS-04 is 120 x 120 x 56 mm), 1.37 kg vs 1.42 kg, 10:1 vs 9:1. The output flange and mounting-hole patterns differ (page gives no hole pattern, so take it from Damiao's drawing), so the printed hip frame and the leg-link flange must be redrawn. Protocol changes from RobStride's CAN protocol to Damiao's CAN protocol (MIT, speed and position modes at 1 Mbps; UART at 921600 bps for setup), so the joint driver code must be rewritten. Supply is listed as 24-48 V (the RS-04 is 15-60 V), so check the pack's full-charge voltage. Connectors are the same types: XT30 power, GH1.25 CAN. checked 2026-10-27 CHANGES AIFIT Lab DM-J10010L-2EC (Include Driver / NOT Include) · $265.00 (motor with integrated driver); $280.00 with USB-to-CAN · Backordered Same as the openelab DM-J10010L-2EC entry: new hip frame and flange, Damiao CAN protocol in place of RobStride's, 24-48 V supply, rated speed 100 rpm vs 167 rpm at 48 V. checked 2026-10-27 CHANGES CubeMars AKE90-8 KV35 · $483.90 (actuator only; driver board not included) A stronger QDD (48 V, 55 N·m rated / 170 N·m peak, 120 rpm rated, 210 rpm no-load, 8:1), 1400 g, round 107.5 mm dia x 43.5 mm. CubeMars' comparison table lists 'Driver Board: No', so a separate driver (page downloads list an AK80-4830-1C-A4 driver; its price was not seen) must be bought, mounted and wired. That means a new hip frame and flange plus room for the driver, and CubeMars' CAN protocol in place of RobStride's in firmware. It costs about 1.9x the RS-04 before the driver. checked 2026-10-27 CHANGES rcdrone.top RobStride-03-1 (RobStride 03) · $224.00 (Motor); $239.00 with USB-to-CAN · Orderable Same maker and same RobStride CAN protocol, so firmware changes the least (motor type and limits only). But it has half the torque: 20 N·m rated / 60 N·m peak, 195 rpm no-load. It is 106 x 106 x 56 mm and 880 g, with flange holes listed as '30.35 x 9 x 6 mm (M4)' and installation holes '98 x 8 x 8 mm (M4)' (the RS-04 lists M5 '35.86 x 9 x 9' and M4 '106 x 10 x 6'), so the hip frame and flange must be redrawn. Only usable if the leg torque budget allows it or another reduction stage is added. checked 2026-10-27

Part details: Uses and sourcing notes leg ×2 the two leg motors, in the hip frame CHECK STOCK Five sellers, all shipping from China; aifitlab and roboticscenter.ai are on backorder (4-6 weeks). Seeed had 11 in stock and 80 due Nov 14 on 2026-10-25. (2026-10-27)

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/odrive-botwheel	ODrive Robotics BotWheel (BW0124) odrive-botwheel	BotWheel (BW0124) ODrive Robotics shop	1 buy 1 × 2-pack	$134.50 $269.00 per 2-pack	$269.00 used $134.50	See note below	Uses and sourcing notes leg ×1 the wheel, on the wheel link's hub CHECK STOCK Only ODrive sells it, in pairs ($269); singles are unavailable. Its EU store showed 40 pairs in stock. Buy spares with the first order. (2026-10-27)	☐	☐
Alternates: 3 alternates · 2 drop-in DROP-IN ODrive Robotics worldwide storefront BW0124_x2 · $269.00 per 2-pack (Single BW0124 $139.00 listed but unavailable) · Orderable; 2-packs only checked 2026-10-27 DROP-IN ODrive Europe BW0124_x2 · EUR 269.00 per pair · In stock (40) checked 2026-10-27 CHANGES AIFIT Lab ZLLG65ASM250-4096 V2.0 (Single Shaft / Not Include Driver) · $120.00 (motor only); $260.00 with ZLAC8015 driver; $390.00 two motors with ZLAC8015D · Backordered The mount changes completely. It has a single-sided cantilever axle with an M16x1.5 thread and 12 mm flats (43 mm long), held by a nut through a slotted plate, instead of the BotWheel flange taking 4 x M6 on a 36 mm circle, so the wheel mount at the foot of the leg must be redrawn. Tyre is 173 ±2 mm dia x 44 mm wide (BotWheel 171 mm), overall width 118.2 mm. Mass is 3.75 kg vs 2.2 kg, i.e. +1.55 kg at each leg tip, which eats into the leg torque budget and needs the balance controller retuned. It has a 4096-line encoder and 15 pole pairs (BotWheel: hall + 3200 CPR), and ZLTECH power and signal connectors instead of JST PUD, so the ODrive harness must be re-terminated and its encoder and pole-pair settings changed; the alternative is ZLTECH's ZLAC8015D driver, which needs different firmware. It is 24 V, 6 N·m rated / 18 N·m peak, 160 rpm rated, IP65. checked 2026-10-27									
☐	buy/lumenier-razor-pro-f4-45a-am32	Lumenier Razor Pro F4 45A 2-6S AM32 single ESC (the flywheel ESC's backup) lumenier-razor-pro-f4-45a-am32	Razor Pro F4 45A 2-6S AM32 ESC (single) seller's no. 22300 (UPC 724376363650) RaceDayQuads available: true (Shopify; no count exposed) (2026-10-27)	2	$31.49	$62.98	1 alternate · 1 drop-in DROP-IN GetFPV not read · $31.49 · unknown (page returns 403 to automated reads) checked 2026-10-27	Uses and sourcing notes turret ×2 esc_backup_pair	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/heatshrink_kit	Heat-shrink assortment heatshrink_kit	Heat Shrink Tube Set (530PC) seller's no. Heat Shrink Tube Set (530PC) RaceDayQuads	1	$9.99	$9.99	1 alternate · 1 drop-in DROP-IN SparkFun PRT-09353 · $10.95 checked 2026-10-27	Uses and sourcing notes electronics ×1	☐	☐
☐	buy/xt60_pair	XT60H connector xt60_pair	Amass XT60H (male + female) seller's no. Amass XT60H (male + female) RaceDayQuads	4	$0.990	$3.96	1 alternate · 1 drop-in DROP-IN Amazon ASIN B0863HJZHB (XT60H pairs) · $1.80 checked 2026-10-27	Uses and sourcing notes electronics ×4 the pack lead (female) and the motors' lead (male)	☐	☐
☐	buy/xt30_male	XT30 connector, male xt30_male	Amass XT30 (male) seller's no. Amass XT30 (male) RaceDayQuads	3	$0.990	$2.97	1 alternate · 1 drop-in DROP-IN GetFPV 5071 (XT30 Power Connectors, 5 Pair) · $3.00 / 5 · In Stock checked 2026-10-27	Uses and sourcing notes electronics ×3 the Y-splice to the two motors' XT30 (female) leads, one spare	☐	☐
☐	buy/tool-am32-usb-linker	USB linker / ESC programming card for AM32 and BLHeli_32 ESCs (tool) tool-am32-usb-linker	VXB0FNQW93SR (Shopify variant 48952943968491) seller's no. VXB0FNQW93SR (Shopify variant 48952943968491) VXB available: true (2026-10-27)	1	$49.99	$49.99	1 alternate · 1 drop-in DROP-IN Amazon ASIN B0FNQW93SR · not shown to this reader checked 2026-10-27	Uses and sourcing notes tools ×1 tool	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/vxb-6810-2rs	6810-2RS Bearing 50x65x7mm - Sealed Ball Bearing - Thin Section (chrome steel) vxb-6810-2rs	Kit7458 (6810-2RS) VXB Bearings	1	$24.99	$24.99	2 alternates · 2 drop-in DROP-IN McMaster-Carr 6656K229 · $33.29 · Delivers tomorrow 7-9 am checked 2026-10-27 DROP-IN Bearings Direct 6810-2RS · $29.19 · In stock - 1 available; usually ships in 24 hours checked 2026-10-27	Uses and sourcing notes turret ×1 pan_bearing	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/feetech-sts3215-c018-12v	Feetech (Waveshare ST3215) STS3215 / ST3215 12 V 30 kg-cm serial bus servo feetech-sts3215-c018-12v	STS3215 / ST3215 12 V 30 kg-cm serial bus servo seller's no. 22414 Waveshare schema.org availability InStock; SKU 22414 'unsaleable': false (no count shown) (2026-10-27)	3	$21.99	$65.97	See note below	See note below	☐	☐
Alternates: 13 alternates · 11 drop-in DROP-IN OpenELAB Feetech-STS3215-C018 · $27.99 · Orderable; ships from China checked 2026-10-27 DROP-IN AIFIT Lab ST-3215-C018 (12V30KG) · $26.00 · Backordered checked 2026-10-27 CHANGES OpenELAB Feetech-STS3250-C002 · $73.99 · Orderable Same Feetech STS serial-bus protocol and 3-pin TTL connector, so firmware stays the same apart from limits. It is stronger (50 kg·cm stall at 12 V, 345:1, 6-12 V) and costs about 2.6x. The page gives no case size or spline, and it says 'a matching TTL interface alone does not establish mechanical compatibility', so check the case and horn against the 45.2 x 24.7 x 35 mm STS3215 before reusing the turret brackets. checked 2026-10-27 CHANGES ROBOTIS America 902-0135-000 (DYNAMIXEL XL430-W250-T) · $27.50 · Orderable The case is 28.5 x 46.5 x 34 mm with a different horn and mounting holes, so the pan/tilt brackets must be redrawn. It speaks DYNAMIXEL protocol over TTL instead of Feetech STS, so the servo driver/SDK and ID setup change. It is weaker: the page gives an estimated rated torque of 0.28 N·m, against about 0.98 N·m (10 kg·cm) rated for the STS3215-C018. Only worth it if the turret load is light. checked 2026-10-27 DROP-IN Amazon ASIN B0CFY52HVV · $30.99 · In Stock (quantity selector to 20+) checked 2026-10-27 DROP-IN Amazon ASIN B0B6P9271Q · $30.90 · Only 1 left in stock - order soon checked 2026-10-27 DROP-IN Amazon ASIN B0FMXVR77Y · $48.99 / 2 · orderable: Add to Cart with a quantity selector (1-8+); the availability line is blank checked 2026-10-27 DROP-IN Amazon ASIN B0CFY52HVV · $30.99 · In Stock (quantity selector to 20+) checked 2026-10-27 DROP-IN Amazon ASIN B0B6P9271Q · $30.90 · Only 1 left in stock - order soon checked 2026-10-27 DROP-IN Amazon ASIN B0FMXVR77Y · $48.99 / 2 · orderable: Add to Cart with a quantity selector (1-8+); the availability line is blank checked 2026-10-27 DROP-IN Amazon ASIN B0CFY52HVV · $30.99 · In Stock (quantity selector to 20+) checked 2026-10-27 DROP-IN Amazon ASIN B0B6P9271Q · $30.90 · Only 1 left in stock - order soon checked 2026-10-27 DROP-IN Amazon ASIN B0FMXVR77Y · $48.99 / 2 · orderable: Add to Cart with a quantity selector (1-8+); the availability line is blank checked 2026-10-27 **Part details:** Uses and sourcing notes turret ×1 pan_servo turret ×1 tilt_servo turret ×1 feed_motor CHECK STOCK Single maker; only Waveshare direct is reliable (ships from China, 3-8 working days express): order the three on day one. US resellers at $28-32 (Amazon, OpenELAB, RobotShop backorder). (2026-10-27)									
☐	buy/waveshare-bus-servo-adapter-a	USB / UART adapter for Feetech serial bus servos waveshare-bus-servo-adapter-a	Bus Servo Adapter (A), SKU 25514 seller's no. 25514 (MPN 'Bus Servo Adapter (A)') Waveshare schema.org availability InStock (2026-10-27)	1	$4.99	$4.99	1 alternate · 1 drop-in DROP-IN Amazon ASIN B0CJ6TP3TP · $12.99 · In Stock checked 2026-10-27	Uses and sourcing notes tools ×1 tool	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/repeat-robotics-am32-35a-esc	Repeat Robotics (sold by Palm Beach Bots) AM32 Brushless Drive ESC 35 A (bi-directional) repeat-robotics-am32-35a-esc	AM32 Brushless Drive ESC 35 A (bi-directional) Palm Beach Bots	2	$24.99	$49.98	See note below	Uses and sourcing notes turret ×1 esc_l turret ×1 esc_r	☐	☐
Alternates: 2 alternates · 1 drop-in DROP-IN Repeat Robotics RR-03-3340 (AM32 35a Single Drive ESC) · $25.00 · Orderable checked 2026-10-27 CHANGES RaceDayQuads 22300 (Lumenier Razor Pro F4 45A 2-6S AM32 ESC) · $31.49 · Orderable At 12 x 28 x 5 mm (3 g without motor wires) it is 2-3 mm longer than the Repeat board's 25-26 x 13 mm, so check the ESC pocket. It is rated 45 A continuous / 60 A burst, 2-6S, no BEC. The page lists DShot1200, Proshot, Oneshot125, Oneshot42 and Multishot input with auto-detect but does not list standard servo PWM, so confirm the controller's output protocol. It is not pre-tuned: set direction, bidirectional mode and throttle limits in the AM32 configurator. It ships with a 220 µF 35 V capacitor that needs a place in the wiring. checked 2026-10-27									
☐	buy/xt60e_f_panel	XT60 female panel socket (the motors' outlet) xt60e_f_panel	Amass XT60E-F seller's no. Amass XT60E-F Jameco Electronics	1	$0.950	$0.95	See note below	Uses and sourcing notes electronics ×1 the pod's outlet: the motors' lead carries the male	☐	☐
Alternates: 3 alternates · 3 drop-in DROP-IN Maplin XT60E-F (Maplin SKU 486711) · Out of stock checked 2026-10-27 DROP-IN Electrokit XT60E-F (Electrokit art. 4103 7928) · Out of stock checked 2026-10-27 DROP-IN LCSC XT60E-M.G.Y (Amass male panel-mount version; LCSC C3040679) · $0.61 · In stock (3,091) checked 2026-10-27										

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/bearing_61812	Steel Ball Bearing, Steel, Shielded, Trade Number 6812-2Z, ABEC Rating 1 — 6812-2RS (Bearings Direct) bearing_61812	6812-2RS Bearings Direct In stock (5) (2026-10-27)	4	$38.58	$154.32	See note below	Uses and sourcing notes leg ×4 support bearings round the two output adapters; two at the apex on the wheel link's hub the plan's seller (docs/leg-prototype-plan.md 6.1, Primary) instead of the catalogue's McMaster-Carr 6656K236	☐	☐
Alternates: 3 alternates · 3 drop-in DROP-IN McMaster-Carr (the catalogue's) 6656K236 · $69.11 each checked 2026-10-26 DROP-IN VXB Bearings Kit7454 · $49.99 · In stock checked 2026-10-27 DROP-IN McMaster-Carr 6656K148 · $64.92 each · Delivers tomorrow 7-9 am Open (no shields): the same 60 x 78 x 10 seat, but nothing keeps the grease in or grit out. Fine in the hip frame; at the apex, next to the wheel, prefer the shielded one. checked 2026-10-27										
☐	buy/bearing_mr126	Corrosion-Resistant 440C Stainless Steel Ball Bearing, Shielded, Trade No. 126-2Z, for 6 mm Shaft Diameter — MR126-2RS (Bearings Direct) bearing_mr126	MR126-2RS Bearings Direct In stock - 646 available (2026-10-27)	1	$4.56	$4.56	See note below	Uses and sourcing notes turret ×1 tilt_bearing the plan's seller (docs/turret-prototype-plan.md 3.3, Primary) instead of the catalogue's McMaster-Carr 7804K112	☐	☐
Alternates: 3 alternates · 2 drop-in DROP-IN McMaster-Carr (the catalogue's) 7804K112 · $8.89 each checked 2026-10-27 DROP-IN VXB Bearings MR126-2RS · $7.77 · In stock checked 2026-10-27 CHANGES McMaster-Carr 7804K142 · $9.08 each · Delivers tomorrow 7-9 am The flanged 126-2Z (flange Ø13.6 x 0.8): the yoke's Ø12 x 4 pocket needs a Ø14 x 0.8 counterbore, or the flange stands 0.8 mm proud of the arm's inner face. checked 2026-10-27										

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/bearing_6301	Deep groove ball bearing 6301-2RSH (12 x 37 x 12, sealed both sides); McMaster-Carr 5972K86 is the same trade number (6301-2RS) bearing_6301	6301-2RSH SKF distributor	4	$12.53	$50.12	NO ALTERNATE	Uses and sourcing notes leg ×4 two in each knee	☐	☐
☐	buy/dewalt-dcb609	DeWalt DCB609 FLEXVOLT 20V/60V MAX 9.0Ah dewalt-dcb609	DCB609 FLEXVOLT 20V/60V MAX 9.0Ah The Power Tool Store	3	$259.00	$777.00	1 alternate · 1 drop-in DROP-IN Ace Tool product page · $259.00 · listed checked 2026-10-27	Uses and sourcing notes stand ×1 bench p0 battery ×2 the battery (D65): two FLEXVOLT 9.0 Ah packs in 20 V mode, in series (10S, 30-41 V); the user's own packs and chargers serve	☐	☐
☐	buy/bracket_corner_8	Corner bracket, 8 series, with 2 x M8 screws + 2 drop-in spring nuts bracket_corner_8	HBLTS8-C-SEP seller's no. HBLTS8-C-SEP MISUMI Ship Date Same Day (2026-10-27)	24	$4.42	$106.08	2 alternates · 2 drop-in DROP-IN MISUMI HBLTS8-C-SEC · $5.24 checked 2026-10-27 DROP-IN 80/20 Inc. 40-4376 + 3 x 75-3422 · $10.90 checked 2026-10-27	Uses and sourcing notes rig ×24 the base's 16 free joint faces and the post's foot (4); 4 spare	☐	☐
☐	buy/extrusion_8080_1200	Aluminium extrusion 8080, 8 series (10 mm slot), cut to 1200 mm extrusion_8080_1200	HFS8-8080-1200 seller's no. HFS8-8080-1200 MISUMI Ship Date Wed. Oct 28, 2026 (2026-10-27)	1	$97.44	$97.44	1 alternate · 1 drop-in DROP-IN 80/20 Inc. 40-8080 cut to 1200 mm · $151.73 checked 2026-10-27	Uses and sourcing notes rig ×1 the post	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/extrusion_4040_620	Aluminium extrusion 4040, 8 series (10 mm slot), cut to 620 mm extrusion_4040_620	HFS8-4040-620 seller's no. HFS8-4040-620 MISUMI Ship Date Wed. Oct 28, 2026 (2026-10-27)	4	$19.09	$76.36	1 alternate · 1 drop-in DROP-IN 80/20 Inc. 40-4040 cut to 620 mm · $30.34 checked 2026-10-27	Uses and sourcing notes rig ×4 the base's cross members: front, wheel, bumper, rear	☐	☐
☐	buy/extrusion_4040_900	Aluminium extrusion 4040, 8 series (10 mm slot), cut to 900 mm extrusion_4040_900	HFS8-4040-900 seller's no. HFS8-4040-900 MISUMI Ship Date Wed. Oct 28, 2026 (2026-10-27)	2	$27.72	$55.44	1 alternate · 1 drop-in DROP-IN 80/20 Inc. 40-4040 cut to 900 mm · $42.69 checked 2026-10-27	Uses and sourcing notes rig ×2 the base's side members	☐	☐
☐	buy/bracket_free_angle_8	Free-angle bracket for 40 mm 8-series profile, with its mounting screws and nuts bracket_free_angle_8	HBLTBA40-SET1 seller's no. HBLTBA40-SET1 MISUMI Ship Date Same Day (2026-10-27)	4	$8.78	$35.12	2 alternates · 2 drop-in DROP-IN MISUMI HBLTBBN40-SET · $10.15 checked 2026-10-27 DROP-IN 80/20 Inc. 40-2085 hinge · $14.34 checked 2026-10-27	Uses and sourcing notes rig ×4 the brace's two ends, two per end (both side faces): up to 2.9 kN in the brace in a jump	☐	☐
☐	buy/extrusion_4080_620	Aluminium extrusion 4080, 8 series (10 mm slot), cut to 620 mm extrusion_4080_620	HFS8-4080-620 seller's no. HFS8-4080-620 MISUMI Ship Date Wed. Oct 28, 2026 (2026-10-27)	1	$34.65	$34.65	1 alternate · 1 drop-in DROP-IN 80/20 Inc. 40-4080 cut to 620 mm · $54.69 checked 2026-10-27	Uses and sourcing notes rig ×1 the post's member, lying flat under it	☐	☐
☐	buy/nut_m4_dropin	Drop-in nut, 8 series, M4 nut_m4_dropin	HNTA8-4 seller's no. HNTA8-4 MISUMI Ship Date Same Day (2026-10-27)	36	$0.900	$32.40	1 alternate · 1 drop-in DROP-IN MISUMI PACK-HNTA8-4 (100) · $62.00 / 100 checked 2026-10-27	Uses and sourcing notes rig ×36 the rails: 17 per rail at 60 mm, 2 spare	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/extrusion_4040_720	Aluminium extrusion 4040, 8 series (10 mm slot), cut to 720 mm extrusion_4040_720	HFS8-4040-720 seller's no. HFS8-4040-720 MISUMI Ship Date Wed. Oct 28, 2026 (2026-10-27)	1	$22.17	$22.17	1 alternate · 1 drop-in DROP-IN 80/20 Inc. 40-4040 cut to 720 mm · $34.75 checked 2026-10-27	Uses and sourcing notes rig ×1 the brace	☐	☐
☐	buy/nut_m8_dropin	Drop-in nut, 8 series, M8 nut_m8_dropin	HNTA8-8 seller's no. HNTA8-8 MISUMI Ship Date Same Day (2026-10-27)	16	$0.900	$14.40	1 alternate · 1 drop-in DROP-IN 80/20 Inc. 75-3422 (M8 x 16 BHSCS + slide-in nut) · $1.14 checked 2026-10-27	Uses and sourcing notes rig ×16 top stop (4), drop bracket (2), pod (4), spares	☐	☐
☐	buy/nut_m6_dropin	Drop-in nut, 8 series, M6 nut_m6_dropin	HNTA8-6 seller's no. HNTA8-6 MISUMI Ship Date Same Day (2026-10-27)	12	$0.900	$10.80	1 alternate · 1 drop-in DROP-IN Fastenal 0961750 (80/20 Inc 13129) · $2.83 · Available Inventory; ships from the manufacturer within 9 days checked 2026-10-27	Uses and sourcing notes rig ×12 the deck (8) and the bottom bumpers (4)	☐	☐
☐	buy/rail_hgr15_kit	HGR15 rail 1000 mm x 2 + HGH15CA block x 4 (one pack) rail_hgr15_kit	ASIN B0GJZCLXWJ (1000mm / 2) seller's no. ASIN B0GJZCLXWJ (1000mm / 2) Amazon Only 12 left in stock (2026-10-27)	1	$69.99	$69.99	See note below	Uses and sourcing notes rig ×1 the carriage guide: two rails on the post's front slots, two blocks each	☐	☐
Alternates: 3 alternates · 3 drop-in DROP-IN Amazon (Coavoo Transmission) ASIN B0BMWQ5J7T · $72.99 · Only 3 left in stock checked 2026-10-27 DROP-IN VXB Bearings VXB0BMWQ5J7T · $159.95 checked 2026-10-27 DROP-IN Amazon (Mycncshop) ASIN B087ZCH3BW (FBT HGR15CA-L1000mm) · $151.00 checked 2026-10-27										

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/terrafirma-dewalt-20v-adapter-10awg	Terrafirma Technology DeWalt 20 V battery adapter, 10 AWG leads (1 ft), 30 A stated, no fuse or switch, 4-pack terrafirma-dewalt-20v-adapter-10awg	4pack 10AWG Adapter for DeWALT 20V (Amazon B07HYJ628S) Amazon listed (2026-10-27)	1 buy 1 pack of 4	$16.00 $64.00 per 4	$64.00 used $16.00	1 alternate · 1 drop-in DROP-IN Amazon (Terrafirma, 12 AWG single) product page · $19.00 · listed; 12 AWG, M4/#6 mounting holes, current not stated checked 2026-10-27	Uses and sourcing notes p0 battery ×1 the packs' rail adapters (4-pack: two used, two spare): 10 AWG leads, 30 A stated	☐	☐
☐	buy/rubber_mat	Rubber mat 3 x 4 ft x 5/16 in, 2-pack rubber_mat	ASIN B0F8RCXB7D seller's no. ASIN B0F8RCXB7D Amazon	1	$52.00	$52.00	1 alternate · 1 drop-in DROP-IN Amazon (uxcell) ASIN B0DP9TXB59 (M8 levelling feet, 8) · $34.00 / 8 checked 2026-10-27	Uses and sourcing notes rig ×1 under the base: the floor carries every member (the wheel member under the tyre most)	☐	☐
☐	buy/sling_1in	Lift-All Type 5 endless polyester sling 1 in x 5 ft (2400 lb vertical) sling_1in	ASIN B0026GIUMG seller's no. ASIN B0026GIUMG Amazon	2	$17.08	$34.16	1 alternate · 1 drop-in DROP-IN Amazon (Kivomot) ASIN B0FRRVHXTR (2-pack, eye-eye 6.5 ft) · $12.58 / 2 checked 2026-10-27	Uses and sourcing notes rig ×2 the tethers: the carriage to the post (girth-hitched under the top stop), the wheel link to the carriage	☐	☐
☐	buy/carabiner	Screwgate carabiner, 25 kN, 2-pack carabiner	ASIN B07D73RRH5 (2-pack) seller's no. ASIN B07D73RRH5 (2-pack) Amazon	4 buy 2 packs of 2	$8.50 $16.99 per 2	$33.98	1 alternate · 1 drop-in DROP-IN Amazon (STURME) ASIN B085HLB374 (2-pack) · $16.99 / 2 checked 2026-10-27	Uses and sourcing notes rig ×4 the tethers' ends	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/dock_adapter_20v	DeWalt 20 V MAX dock adapter, 12 AWG leads, switch and 30 A blade fuse dock_adapter_20v	ASIN B0B9NPZM3M seller's no. ASIN B0B9NPZM3M Amazon In Stock (2026-10-27)	2	$9.99	$19.98	1 alternate · 1 drop-in DROP-IN Amazon (Ensayeer, sold by Unsaidtech) ASIN B09J4R4QZ6 (2-pack) · $22.00 / 2 · In Stock checked 2026-10-27	Uses and sourcing notes electronics ×2 D65: one per DCB609 in 20 V mode; the two in series (10S, 30-41 V) feed the pod	☐	☐
☐	buy/xt90s_pair	Amass XT90-S anti-spark connector, male + female, 5 pairs xt90s_pair	XT90-S Sky RC Hobby on Amazon In Stock (ships from Amazon) (2026-10-27)	2 buy 1 pack of 5	$2.60 $12.98 per 5	$12.98 used $5.19	1 alternate · 1 drop-in DROP-IN FLYRC on Amazon product page · $14.99 / 3 · In Stock; 3 pairs with 150 mm 10 AWG pigtails checked 2026-10-27	Uses and sourcing notes p0 battery ×2 the bay's lead (anti-spark: female on the bay, male on the robot) and the series link between the packs	☐	☐
☐	buy/load_cells_4x50	4 x 50 kg half-bridge load cells + HX711 load_cells_4x50	ASIN B079FTXR7Y seller's no. ASIN B079FTXR7Y Amazon Only 10 left (2026-10-27)	1	$7.90	$7.90	1 alternate · 1 drop-in DROP-IN VXB VXB07X87GYLQ · $39.99 checked 2026-10-27	Uses and sourcing notes electronics ×1 the load plate under the wheel (not in jumps or drops)	☐	☐
☐	buy/traco-tep-150-4812wi	TRACO TEP 150-4812WI DC-DC converter, 18-75 V in (4:1), 12 V 12.5 A (150 W), isolated 2,250 V DC, chassis mount, screw terminals traco-tep-150-4812wi	TEP 150-4812WI Newark 4 (as an aggregator, electronicsdatasheets.com, lists it: confirm on the order) (2026-10-27)	1	$275.99	$275.99	1 alternate · 1 drop-in DROP-IN Farnell product page · $222.65 · listed (via the same aggregator) checked 2026-10-27	Uses and sourcing notes p0 power skid ×1 12 V 12.5 A, 18-75 V in, isolated (D65): the J501 and the 12 V rail, in place of the DDR-120C-24 and the DDR-60L-12 (both start too high or are too small for 30-41 V)	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/tvs-p6ke51a	TVS diode 600 W unidirectional, 43.6 V standoff, 70.1 V clamp, DO-15 (P6KE51A) tvs-p6ke51a	P6KE51A seller's no. 99AC4212 Newark 1476 (2026-10-27)	3	$1.28	$3.84	1 alternate · 1 drop-in DROP-IN TME P6KE51A-DIO · $0.47 · 4628 checked 2026-10-27	Uses and sourcing notes turret ×3 el_d_tvs51	☐	☐
☐	buy/yageo-mfr-25fbf52-10r	Resistor 10.0 R 1 % 1/4 W metal film (Yageo MFR-25FBF52) yageo-mfr-25fbf52-10r	MFR-25FBF52-10R seller's no. 49AK5972 Newark 9608 (2026-10-27)	2	$0.100	$0.20	1 alternate · 1 drop-in DROP-IN Quest Components product page · $0.07 · 88 checked 2026-10-27	Uses and sourcing notes turret ×2 el_r_10r	☐	☐
☐	buy/remington-18ptfestr-25ft	Hook-up wire 18 AWG stranded PTFE, 600 V, 25 ft remington-18ptfestr-25ft	18PTFESTRBLA25 (black) / 18PTFESTRRED25 (red) Digi-Key	1 buy 1 pack of 25	$2.58 $64.56 per 25	$64.56 used $2.58	1 alternate · 1 drop-in DROP-IN Mouser 602-6825-100-02 · $120.87 / 100 · 2 (Mouser, via oemstrade feed today) checked 2026-10-27	Uses and sourcing notes turret ×1 el_wire_18awg	☐	☐
☐	buy/tdk-lambda-i6a4w020a033v-001-r	i6A4W 250 W non-isolated buck, 9-53 V in, 3.3-15 V out by one trim resistor, 20 A (1/16 brick, horizontal through-hole) tdk-lambda-i6a4w020a033v-001-r	i6A4W020A033V-001-R seller's no. I6A4W020A033V-001-R Digi-Key 1873 (2026-10-27)	1	$47.46	$47.46	1 alternate · 1 drop-in DROP-IN Newark 81Y1493 · $40.32 · 36 in one snapshot, back-order in another checked 2026-10-27	Uses and sourcing notes turret ×1 el_flywheel_regulator	☐	☐
☐	buy/quarton-vlm-650-03-lpt	Red dot laser module 650 nm, < 1 mW (Class 2), 2.6-6 V, Ø7 x 22.5 mm quarton-vlm-650-03-lpt	VLM-650-03 LPT seller's no. VLM-650-03-LPT-ND Digi-Key 966 (authorized; Octopart data updated 2026-10-26 12:48 UTC) (2026-10-27)	1	$25.03	$25.03	See note below	Uses and sourcing notes turret ×1 laser	☐	☐
Alternates: 3 alternates · 1 drop-in CHANGES Digi-Key Quarton VLM-650-28 LPT · 63 (snippet) check its diameter and length against the Ø7 bore before ordering checked 2026-10-27 CHANGES Digi-Key Quarton VLM-520-73 LPT (green, Class 2) · $43.49 · 189 (Octopart 2026-10-26) Ø7 x 26.1: 5 mm longer than the red one; the mount holds it as drawn for v1 checked 2026-10-26 DROP-IN Digi-Key VLM-635-03LPT-ND · $28.91 · 36 (authorized; Octopart data updated 2026-10-26 12:48 UTC) checked 2026-10-27										

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/infineon-irlz44npbf	Logic-level N-MOSFET 55 V 47 A, TO-220 infineon-irlz44npbf	IRLZ44NPBF seller's no. IRLZ44NPBF-ND Digi-Key	6	$1.80	$10.80	1 alternate · 1 drop-in DROP-IN Mouser 942-IRLZ44NPBF · $1.80 · 77849 checked 2026-10-27	Uses and sourcing notes turret ×6 el_brake_fets	☐	☐
☐	buy/te-dt06-12sa-kit	CONN PLUG HSG 12POS - DT 12-way plug (socket side), key A te-dt06-12sa-kit	DT06-12SA (+ W12S wedge + 12 x 0462-201-16141 sockets) Digi-Key	1	$8.50	$8.50	See note below	Uses and sourcing notes turret ×1 dt_plug	☐	☐
Alternates: 1 alternate · 1 drop-in DROP-IN Chief Enterprises AT06-12SA plug + AW12S wedge + 12 x AT62-201-16141 sockets (Amphenol Sine Systems) · AT06-12SA $3.22 (1-9); AW12S $0.31 (1-9); AT62-201-16141 $0.58 (1-9). Kit = $3.22 + $0.31 + 12 x $0.58 = $10.49 · Orderable None if paired with the Amphenol AT04-12PA-L012 receptacle (Amphenol names AT04-12PA as the mate). Mating with the TE DT04-12PA-L012 relies on AT/DT intermating, which no page I read states. checked 2026-10-27										
☐	buy/stop_relay_dpdt	Omron G2RL-2 DC12 power relay, DPDT, 12 V coil, 8 A at 30 V DC (the stop relay: the turret's safety loop and its 24 V branch) stop_relay_dpdt	G2RL-2-DC12 DigiKey 1,078 immediate (search result, 2026-10-27; the live page to confirm) (2026-10-27)	2	$4.14	$8.28	1 alternate · 1 drop-in DROP-IN LCSC product page · listed as C61458 (price shown on the live page) checked 2026-10-27	Uses and sourcing notes p0 control ×2 the stop relay (one spare): pole A the turret's safety loop, pole B its bus branch (D66); on the MCU board	☐	☐
☐	buy/fuse_holder_5x20	5 x 20 mm panel fuse holder fuse_holder_5x20	Littelfuse 03455LS1H seller's no. Littelfuse 03455LS1H Digi-Key	1	$7.16	$7.16	See note below	Uses and sourcing notes electronics ×1 the 12 V converter's input fuse, on the panel	☐	☐
Alternates: 2 alternates · 2 drop-in DROP-IN LCSC 03455LS1HX020 (Littelfuse; LCSC C206937) · $1.40 · Out of stock checked 2026-10-27 DROP-IN Schurter 0031.1756 (FEU = body 0031.1673 + fingergrip cap 0031.1741, 5x20 mm, solder, 90 deg) checked 2026-10-27										

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/fuse-holder-5x20-panel	FUSE HLDR CART 250V 10A PNL MNT - 345 series shock-safe panel holder, 5 x 20 mm, low-profile body, screwdriver-slot knob, 3/16 in right-angle dual-purpose solder/QC terminals fuse-holder-5x20-panel	03455LS1H Digi-Key	1	$7.16	$7.16	See note below	See note below	☐	☐
Alternates: 2 alternates · none drop-in CHANGES LCSC 03455LS1HX020 (Littelfuse; LCSC C206937) · $1.3968 (1+), $1.1671 (10+), $1.0407 (30+), $0.8973 (100+), $0.8341 (500+), $0.8048 (1000+) · Out of stock Same 345-series holder as the primary (the page does not explain the X020 suffix), so it needs the same 12.70-12.83 mm hole with a flat, not a 12.6 mm round hole. checked 2026-10-27 CHANGES Schurter 0031.1756 (FEU = body 0031.1673 + fingergrip cap 0031.1741, 5x20 mm, solder, 90 deg) M12.7 x 1.5 thread (nut SW14). The datasheet's mounting-hole figures are 12.7 (+0.1/0) and 13 (-0.3) mm round, with flat options 12.2 (-0.13) and 11.7 (-0.13) mm (read as text; check the drawing). The 12.6 mm hole is too small; open it to about 12.8-13.0 mm. Panel thickness max 7 mm, so the 4 mm wall is fine. Caps are ordered separately unless the complete code is used. checked 2026-10-27 **Part details:** Uses and sourcing notes stand ×1 bench DOESN'T FIT AS DRAWN The Littelfuse 345 needs a 12.70-12.83 mm hole with an anti-rotation flat (12.07-12.19 across); Schurter's FEU 12.7-13. The stand's 12.6 mm round hole is too small. The 4 mm wall is fine. (2026-10-27)									
☐	buy/te-0460-202-16141	DT size 16 pin contact, nickel, 14-16 AWG te-0460-202-16141	0460-202-16141 seller's no. 1734-1328-ND Digi-Key	12	$0.450	$5.40	1 alternate · 1 drop-in DROP-IN Mouser 571-0460-202-16141 · $0.45 · 267161 checked 2026-10-27	Uses and sourcing notes turret ×12 el_dt_pins	☐	☐
☐	buy/littelfuse-0477005-mxp	5 x 20 mm time-lag fuse, 5 A, 500 V AC / 400 V DC (477 series) littelfuse-0477005-mxp	Littelfuse 0477005.MXP seller's no. Littelfuse 0477005.MXP Digi-Key	4	$1.00	$4.00	2 alternates · 2 drop-in DROP-IN Mouser 576-0477005.MXP (Littelfuse 0477005.MXP) · $2.18 · 8590 in stock (Mouser's figure as listed on findchips.com today) checked 2026-10-27 DROP-IN Digi-Key product page checked ?	Uses and sourcing notes electronics ×2 the 12 V converter's input (one spare) stand ×2 bench	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/molex-43031-0001	Micro-Fit 3.0 male crimp terminal, 20-24 AWG, tin molex-43031-0001	43031-0001 seller's no. WM1841CT-ND Digi-Key	40	$0.085	$3.40	1 alternate · 1 drop-in DROP-IN Mouser 538-43031-0001-CT · $8.50 / 100 · 5000 checked 2026-10-27	Uses and sourcing notes turret ×40 el_microfit_m24	☐	☐
☐	buy/panasonic-eeu-fr1j471b	Aluminium electrolytic, 470 uF, 63 V, low ESR (FR), 12.5 x 25 mm radial panasonic-eeu-fr1j471b	EEU-FR1J471B seller's no. P15300CT-ND Digi-Key ~20,107 (findchips row, undated) (2026-10-27)	2	$1.54	$3.08	1 alternate · 1 drop-in DROP-IN Newark 62W8085 · $2.13 · 10970 checked 2026-10-27	Uses and sourcing notes turret ×2 el_capacitors	☐	☐
☐	buy/allegro-a1120lua-t	Unipolar hall-effect switch, 3-24 V, SIP-3 (UA) allegro-a1120lua-t	A1120LUA-T seller's no. 620-1330-ND Digi-Key 22190 (2026-10-27)	1	$1.79	$1.79	1 alternate · 1 drop-in DROP-IN Mouser 250-A1120LUA-T · $1.79 · 2262 checked 2026-10-27	Uses and sourcing notes turret ×1 hall_switch	☐	☐
☐	buy/molex-43020-1200	Micro-Fit 3.0 plug housing, dual row, 12 circuits molex-43020-1200	43020-1200 seller's no. WM1782-ND Digi-Key	2	$0.820	$1.64	1 alternate · 1 drop-in DROP-IN Mouser 538-43020-1200 · $0.82 · 13814 checked 2026-10-27	Uses and sourcing notes turret ×2 el_ring_plugs	☐	☐
☐	buy/molex-43025-1200	Micro-Fit 3.0 receptacle housing, dual row, 12 circuits molex-43025-1200	43025-1200 seller's no. WM1788-ND Digi-Key	2	$0.810	$1.62	1 alternate · 1 drop-in DROP-IN Mouser 538-43025-1200 · $0.81 · 75541 checked 2026-10-27	Uses and sourcing notes turret ×2 el_ring_receptacles	☐	☐
☐	buy/vishay-4n35	Optocoupler, phototransistor output, DIP-6 vishay-4n35	4N35 seller's no. 4N35-ND Digi-Key	1	$1.58	$1.58	1 alternate · 1 drop-in DROP-IN Mouser 859-4N35 · $0.64 · 9390 checked 2026-10-27	Uses and sourcing notes turret ×1 el_opto	☐	☐
☐	buy/bourns-mf-r300	Resettable PTC fuse, 3 A hold, 30 V bourns-mf-r300	MF-R300 seller's no. MF-R300-ND Digi-Key	3	$0.500	$1.50	1 alternate · 1 drop-in DROP-IN Mouser 652-MFR300 · $0.50 · 5353 checked 2026-10-27	Uses and sourcing notes turret ×3 el_polyfuse_3a	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/vishay-teft4300	Silicon NPN phototransistor, T-1 (3 mm) vishay-teft4300	TEFT4300 seller's no. 751-1041-ND Digi-Key	2	$0.700	$1.40	1 alternate · 1 drop-in DROP-IN Mouser 782-TEFT4300 · $0.70 · 2446 checked 2026-10-27	Uses and sourcing notes turret ×2 el_gate_receivers	☐	☐
☐	buy/molex-43030-0038	Micro-Fit 3.0 female crimp terminal, 18 AWG, tin, 8.5 A molex-43030-0038	43030-0038 seller's no. WM13070CT-ND Digi-Key	8	$0.160	$1.28	1 alternate · 1 drop-in DROP-IN Mouser 538-43030-0038-LP · $0.16 · 79499 checked 2026-10-27	Uses and sourcing notes turret ×8 el_microfit_f18	☐	☐
☐	buy/molex-43031-0021	Micro-Fit 3.0 male crimp terminal, 18 AWG, tin, 8.5 A molex-43031-0021	43031-0021 seller's no. WM22521CT-ND Digi-Key	8	$0.160	$1.28	1 alternate · 1 drop-in DROP-IN Mouser 538-43031-0021-LP · $0.16 · 9188 checked 2026-10-27	Uses and sourcing notes turret ×8 el_microfit_m18	☐	☐
☐	buy/onsemi-2n7000	N-channel MOSFET 60 V 200 mA, TO-92 onsemi-2n7000	2N7000 seller's no. 2N7000FS-ND Digi-Key	1	$1.23	$1.23	1 alternate · 1 drop-in DROP-IN Mouser 512-2N7000 · $1.23 · 95079 checked 2026-10-27	Uses and sourcing notes turret ×1 el_laser_switch	☐	☐
☐	buy/ti-sn74hc74n	Dual D flip-flop, DIP-14 ti-sn74hc74n	SN74HC74N seller's no. 296-1602-5-ND Digi-Key 9538 (2026-10-27)	1	$1.13	$1.13	1 alternate · 1 drop-in DROP-IN Mouser 595-SN74HC74N · $1.11 · 10808 checked 2026-10-27	Uses and sourcing notes turret ×1 el_trip_latch	☐	☐
☐	buy/molex-43030-0001	Micro-Fit 3.0 female crimp terminal, 20-24 AWG, tin molex-43030-0001	43030-0001 seller's no. WM1837CT-ND Digi-Key	14	$0.075	$1.05	1 alternate · 1 drop-in DROP-IN Mouser 538-43030-0001-CT · $7.50 / 100 · 257100 checked 2026-10-27	Uses and sourcing notes turret ×14 el_microfit_f24	☐	☐
☐	buy/molex-43020-1000	Micro-Fit 3.0 plug housing, dual row, 10 circuits molex-43020-1000	43020-1000 seller's no. WM1781-ND Digi-Key	1	$0.970	$0.97	1 alternate · 1 drop-in DROP-IN Mouser 538-43020-1000 · $0.97 · 20449 checked 2026-10-27	Uses and sourcing notes turret ×1 el_ct_signal_plug	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/vishay-tsal4400	IR emitting diode 940 nm, T-1 (3 mm) vishay-tsal4400	TSAL4400 seller's no. 751-1201-ND Digi-Key	2	$0.480	$0.96	1 alternate · 1 drop-in DROP-IN Mouser 782-TSAL4400 · $0.48 · 18862 checked 2026-10-27	Uses and sourcing notes turret ×2 el_gate_emitters	☐	☐
☐	buy/ti-sn74ahct125n	Quad buffer, 3-state, TTL inputs, DIP-14 ti-sn74ahct125n	SN74AHCT125N seller's no. 296-4655-5-ND Digi-Key 1674 (2026-10-27)	1	$0.950	$0.95	1 alternate · 1 drop-in DROP-IN Mouser 595-SN74AHCT125N · $0.98 · 3136 checked 2026-10-27	Uses and sourcing notes turret ×1 el_bus_buffer	☐	☐
☐	buy/te-w12p	DT wedgelock for the 12-way receptacle te-w12p	W12P seller's no. 1734-1034-ND Digi-Key	1	$0.880	$0.88	1 alternate · 1 drop-in DROP-IN Mouser 571-W12P-P004 · $0.88 · 2620 checked 2026-10-27	Uses and sourcing notes turret ×1 el_dt_wedge	☐	☐
☐	buy/ti-sn74hc08n	Quad 2-input AND, DIP-14 ti-sn74hc08n	SN74HC08N seller's no. 296-1570-5-ND Digi-Key 22274 (2026-10-27)	1	$0.790	$0.79	1 alternate · 1 drop-in DROP-IN Mouser 595-SN74HC08N · $0.78 · 8674 checked 2026-10-27	Uses and sourcing notes turret ×1 el_and	☐	☐
☐	buy/molex-43025-1000	Micro-Fit 3.0 receptacle housing, dual row, 10 circuits molex-43025-1000	43025-1000 seller's no. WM1787-ND Digi-Key	1	$0.750	$0.75	1 alternate · 1 drop-in DROP-IN Mouser 538-43025-1000 · $0.75 · 131360 checked 2026-10-27	Uses and sourcing notes turret ×1 el_ct_signal_receptacle	☐	☐
☐	buy/ti-ne555p	Precision timer, DIP-8 ti-ne555p	NE555P seller's no. 296-NE555P-ND Digi-Key 28579 (2026-10-27)	1	$0.590	$0.59	1 alternate · 1 drop-in DROP-IN Mouser 595-NE555P · $0.59 · 61909 checked 2026-10-27	Uses and sourcing notes turret ×1 el_555	☐	☐
☐	buy/ti-lm393p	Dual comparator, DIP-8 ti-lm393p	LM393P seller's no. 296-1398-5-ND Digi-Key 8592 (2026-10-27)	1	$0.530	$0.53	1 alternate · 1 drop-in DROP-IN Mouser 595-LM393P · $0.53 · 23356 checked 2026-10-27	Uses and sourcing notes turret ×1 el_gate_comparator	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/molex-43025-0600	Micro-Fit 3.0 receptacle housing, dual row, 6 circuits molex-43025-0600	43025-0600 seller's no. WM1785-ND Digi-Key	1	$0.510	$0.51	1 alternate · 1 drop-in DROP-IN Mouser 538-43025-0600 · $0.51 · 207720 checked 2026-10-27	Uses and sourcing notes turret ×1 el_ct_power_receptacle	☐	☐
☐	buy/bourns-mf-r250	PTC resettable fuse, 30 V, 2.5 A hold, 5.0 A trip, radial bourns-mf-r250	MF-R250 seller's no. MF-R250-ND Digi-Key	1	$0.490	$0.49	1 alternate · 1 drop-in DROP-IN Mouser 652-MFR250 · $0.49 · 8520 checked 2026-10-27	Uses and sourcing notes turret ×1 el_polyfuse_2a	☐	☐
☐	buy/molex-43020-0600	Micro-Fit 3.0 plug housing, dual row, 6 circuits molex-43020-0600	43020-0600 seller's no. WM1779-ND Digi-Key	1	$0.400	$0.40	1 alternate · 1 drop-in DROP-IN Mouser 538-43020-0600 · $0.40 · 12657 checked 2026-10-27	Uses and sourcing notes turret ×1 el_ct_power_plug	☐	☐
☐	buy/caldwell-ballistic-precision-chronograph-bt	Ballistic Precision chronograph with Bluetooth (optical) caldwell-ballistic-precision-chronograph-bt	4001147 seller's no. SKU 4001147 Caldwell Availability: In Stock (2026-10-27)	1	$168.99	$168.99	2 alternates · 2 drop-in DROP-IN Competition Electronics CEI-3820 · $139.95 · available: true checked 2026-10-27 DROP-IN Graf & Sons CE3820 (Shopify variant 46624058671235) · $136.99 · available: true checked 2026-10-27	Uses and sourcing notes stand ×1 bench	☐	☐
☐	buy/pololu-d36v50f12-4095	12V, 4.5A Step-Down Voltage Regulator D36V50F12 pololu-d36v50f12-4095	4095 (D36V50F12) Pololu	2	$39.95	$79.90	1 alternate · 1 drop-in DROP-IN Pololu 5577 · $47.95 · 220 checked 2026-10-27	Uses and sourcing notes turret ×2 el_12v_regulators	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/buck_5v	Pololu 5 V 3.2 A step-down regulator D36V28F5 (from the 12 V rail) buck_5v	D36V28F5 (#3782) Pololu Rationed (active and preferred) (2026-10-27)	1	$23.95	$23.95	1 alternate · 1 drop-in DROP-IN Pololu product page · $26.95 · Rationed; D24V22F5 5 V 2.5 A, 5.3-36 V in checked 2026-10-27	Uses and sourcing notes p0 power skid ×1 5 V from the 12 V rail: the MCU, VN-100, CAN transceivers, radio receiver	☐	☐
☐	buy/pololu-d36v28f5-3782	5V, 3.2A Step-Down Voltage Regulator D36V28F5 pololu-d36v28f5-3782	3782 (D36V28F5) Pololu	1	$23.95	$23.95	1 alternate · none drop-in CHANGES Pololu 2858 · $26.95 · 141 D24V22F5: 2.5 A against 3.2 A, input at most 36 V (fine on 24 V); a different footprint on the level-1 board checked 2026-10-27	Uses and sourcing notes turret ×1 el_5v_regulator	☐	☐
☐	buy/radio_stop_relay	Pololu RC switch with relay (the radio stop: its dry contact is the STOP loop to the MCU; D37 calls this the simplest non-rated version) radio_stop_relay	2804 Pololu listed (the body plan, read 2026-10-27) (2026-10-27)	1	$19.95	$19.95	1 alternate · 1 drop-in DROP-IN Pololu product page · the digital-output version (no relay): backup through a MOSFET on the MCU board checked 2026-10-27	Uses and sourcing notes p0 control ×1 the radio's switch: its contact feeds the stop relay's coil (closed = run, open on the failsafe)	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/pololu-ul1007-24awg-60ft	Stranded hook-up wire, 24 AWG UL1007, 60 ft spool pololu-ul1007-24awg-60ft	2630 (black), 2632 (red), 2634 (yellow), 2639 (white) seller's no. 2630 Pololu InStock (page data-available-stock=441); category page showed some colours as out of stock/backorder (2026-10-27)	4 buy 1 pack of 60	$0.1825 $10.95 per 60	$10.95 used $0.73	1 alternate · 1 drop-in DROP-IN Digi-Key product page · $31.63 / 6 checked 2026-10-27	Uses and sourcing notes turret ×4 el_wire_24awg	☐	☐
☐	buy/pololu-2815	Big MOSFET Slide Switch with Reverse Voltage Protection, HP pololu-2815	2815 Pololu	1	$8.49	$8.49	See note below	Uses and sourcing notes turret ×1 el_safety_switch	☐	☐
Alternates: 1 alternate · none drop-in CHANGES Pololu 2813 · $8.95 · 211 the 2813 Big Pushbutton Power Switch HP: the same HP ratings, but its control is ON/OFF/CTRL pins, not the slide switch's ON pin: the arm chain's ARM output drives CTRL (check Pololu's pin logic before fitting) checked 2026-10-27										
☐	buy/pendant_specs	Momentary switch, 35 mm, 6 ft cord, 3.5 mm mono plug pendant_specs	Specs IC (10000114) seller's no. Specs IC (10000114) AbleNet In stock (2026-10-27)	1	$75.00	$75.00	1 alternate · 1 drop-in DROP-IN AbleNet Jelly Bean (10033400) · $75.00 checked 2026-10-27	Uses and sourcing notes electronics ×1 hold-to-run: the trigger of the off-delay relay and the rig node's input	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/pendant-hold-to-run	Specs IC switch - 1 3/8 in (3.5 cm) momentary button, 3.5 mm mono plug, 6 ft (183 cm) cord pendant-hold-to-run	Specs IC (SKU 10000114) seller's no. Product Number 10000114 (newsletter: 1000114) AbleNet schema.org InStock (structured data; no visible stock text) (2026-10-27)	1	$75.00	$75.00	See note below	Uses and sourcing notes stand ×1 bench	☐	☐
Alternates: 3 alternates · 1 drop-in CHANGES Adafruit Product ID 423 (+ Product ID 2790 plug) · $10.00 · both 'In stock' (423: $7.50; 2790: $2.50) a foot switch (Adafruit 423, momentary, 250 VAC 10 A) wired to an Adafruit 2790 3.5 mm plug: $10 for both checked 2026-10-27 DROP-IN AbleNet Jelly Bean (SKU 10033400, mpn 1.3.0001) · $75.00 · In stock checked 2026-10-27 CHANGES Coyote Direct Motion Raceworks 'Momentary Small Push Button w/6' Wound Cord' (no SKU on page) · $60.00 (Red or Green); $120.00 (Both) · Orderable Ends in two bare 18 AWG wires, so wire on a 3.5 mm mono plug (not included). Sealed 10 A momentary button with a 0.625 in body and a 6 ft coiled cord when pulled tight; lock nut and washer included, no bracket. A true hand-held pendant (transbrake-style button). checked 2026-10-27									
☐	buy/contactor_lev100	Sealed DC contactor 100 A, 900 VDC, 12 V coil (5.5 W) contactor_lev100	TE Kilovac LEV100A4ANG (3-1618389-7) seller's no. TE Kilovac LEV100A4ANG (3-1618389-7) TTI, Inc. 301 In Stock (2026-10-27)	1	$86.64	$86.64	2 alternates · 2 drop-in DROP-IN TTI, Inc. TE EV200AAANA · $127.61 · 439 In Stock checked 2026-10-27 DROP-IN RS Americas LEV100A4ANG (RS 70062417) · $76.47 checked 2026-10-27	Uses and sourcing notes electronics ×1 the hold-to-run power cut in the pack's positive	☐	☐
☐	buy/resistor_10r_25w	10 ohm 25 W aluminium-housed resistor (box of 10) resistor_10r_25w	TE HSA2510RJ seller's no. TE HSA2510RJ TTI, Inc.	1 buy 1 pack of 10	$3.70 $37.00 per 10	$37.00 used $3.70	1 alternate · 1 drop-in DROP-IN Mouser 279-THS2510RJ (TE Connectivity THS2510RJ) · $7.19 · 208 in stock (Mouser's figure as listed on findchips.com today) checked 2026-10-27	Uses and sourcing notes electronics ×1 optional: a 1 s pre-charge across the contactor at the first power-up if it sparks	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/resistor_120	120 ohm 1/4 W (1000) resistor_120	YAGEO CFR-25JB-52-120R seller's no. YAGEO CFR-25JB-52-120R TTI, Inc.	2 buy 1 pack of 1000	$0.0156 $15.60 per 1000	$15.60 used $0.03	1 alternate · 1 drop-in DROP-IN Mouser 603-CFR-25JB-52-120R (YAGEO CFR-25JB-52-120R) · $0.10 · 19198 in stock (Mouser's figure as listed on findchips.com today) checked 2026-10-27	Uses and sourcing notes electronics ×2 CAN terminators where the adapter's or a motor's own is not used	☐	☐
☐	buy/shunt_150a	Blue Sea meter shunt 150 A / 50 mV (manganin), in the battery negative shunt_150a	analog meter shunt 150 A / 50 mV (UPC 632085092313) go2marine available (2026-10-27)	1	$48.74	$48.74	1 alternate · 1 drop-in DROP-IN The Solar Store product page · $59.00 · available; Deltec MKB-100-100 (100 A / 100 mV) checked 2026-10-27	Uses and sourcing notes p0 power skid ×1 the current shunt in the + lead after the MRBF (high side: the INA228 survives the switch opening)	☐	☐
☐	buy/mrbf_80a	Blue Sea 5181 MRBF terminal fuse 80 A, 58 V DC, 2 kA at 58 V mrbf_80a	5181 go2marine available (2026-10-27)	2	$20.50	$41.00	1 alternate · 1 drop-in DROP-IN go2marine product page · $13.50 · available; Egis MRBF 80 A, 58 V DC checked 2026-10-27	Uses and sourcing notes p0 power skid ×2 the main fuse (80 A, 58 V DC, 2 kA at 58 V); one spare	☐	☐
☐	buy/blue-sea-5191-mrbf-terminal-fuse-block	Blue Sea Systems 5191 MRBF Terminal Fuse Block 30-300A blue-sea-5191-mrbf-terminal-fuse-block	5191 MRBF Terminal Fuse Block 30-300A go2marine available (2026-10-27)	1	$31.50	$31.50	1 alternate · 1 drop-in DROP-IN go2marine product page · $17.25 · available; Egis MRBF block, single output, 58 V DC checked 2026-10-27	Uses and sourcing notes p0 power skid ×1 the main fuse's block on the bay's + lead (on 3 mm printed PETG spacers; its M8 post insulated from the plate)	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/adafruit-feather-m4-can-4759	Feather M4 CAN Express with ATSAME51 adafruit-feather-m4-can-4759	4759 Adafruit	1	$24.95	$24.95	2 alternates · 2 drop-in DROP-IN Digi-Key 1528-4759-ND · $24.95 · 473 checked 2026-10-27 DROP-IN Mouser 485-4759 · $24.95 · 455 checked 2026-10-27	Uses and sourcing notes turret ×1 el_mcu	☐	☐
☐	buy/feather_m4_can	Feather M4 CAN Express (ATSAME51) feather_m4_can	4759 seller's no. 4759 Adafruit In stock (2026-10-27)	1	$24.95	$24.95	1 alternate · 1 drop-in DROP-IN Adafruit 5724 RP2040 CAN Bus Feather · $19.95 checked 2026-10-27	Uses and sourcing notes electronics ×1 the rig node: the pendant's loop, RUN frames, the ToF, the load cells, the bus voltage	☐	☐
☐	buy/ina228	Adafruit INA228 85 V 20-bit power monitor breakout (remove its 15 mOhm shunt; wire it to the external shunt) ina228	5832 Adafruit 58 in stock (2026-10-27)	1	$14.95	$14.95	1 alternate · 1 drop-in DROP-IN Adafruit product page · $10.95 · In stock; INA238 (16-bit), same board checked 2026-10-27	Uses and sourcing notes p0 power skid ×1 pack voltage and current to the MCU over I2C (its own 15 mOhm shunt removed, wired to the external shunt)	☐	☐
☐	buy/tof_vl53l1x	VL53L1X time-of-flight breakout, 30-4000 mm tof_vl53l1x	3967 seller's no. 3967 Adafruit In stock (2026-10-27)	1	$14.95	$14.95	1 alternate · 1 drop-in DROP-IN SparkFun SEN-14722 · $29.95 checked 2026-10-27	Uses and sourcing notes electronics ×1 the carriage's height (jump apex) at 50 Hz	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/can_pal	Adafruit CAN Pal (TJA1051T/3) CAN transceiver, switchable 120 Ohm can_pal	5708 Adafruit In stock (2026-10-27)	3	$3.95	$11.85	1 alternate · 1 drop-in DROP-IN Waveshare product page · $4.99 · InStock; SN65HVD230 board checked 2026-10-27	Uses and sourcing notes p0 control ×3 CAN transceivers for the MCU's buses A, B, C	☐	☐
☐	buy/tool-microfit-crimper	Ratcheting crimper for open-barrel contacts, 18-28 AWG (Adafruit 1213): Micro-Fit 3.0 terminals (tool) tool-microfit-crimper	Product ID 1213 seller's no. Product ID 1213 Adafruit 'In stock'; 'On Sale' $10.00 (struck-through $34.95; qty-break table still shows $34.95 for 1-9) (2026-10-27)	1	$10.00	$10.00	2 alternates · 2 drop-in DROP-IN Adafruit Product ID 350 · $49.95 · 'In stock' checked 2026-10-27 DROP-IN Mouser 538-63819-0000 (Molex 63819-0000) · $501.22 · 40 in stock (Mouser); Heilind 42 in stock at $463.32 checked 2026-10-27	Uses and sourcing notes tools ×1 tool	☐	☐
☐	buy/feather wing_proto	FeatherWing Proto featherwing_proto	2884 seller's no. 2884 Adafruit In Stock (2026-10-27)	2	$4.95	$9.90	1 alternate · 1 drop-in DROP-IN Mouser 485-2884 (Adafruit 2884) · $4.95 · 91 in stock (Mouser's figure as listed on findchips.com today) checked 2026-10-27	Uses and sourcing notes electronics ×2 one under the Feather (the dividers, the sensors' plugs); one as the carriage's CAN junction (the GH headers)	☐	☐
☐	buy/ferrules	Wire ferrules, 5000 pcs, AWG 22-7 (every screw terminal) ferrules	5000 pcs AWG 22-7 Haisstronica available (2026-10-27)	1	$65.99	$65.99	1 alternate · 1 drop-in DROP-IN Haisstronica product page · $33.99 · available (the tool) checked 2026-10-27	Uses and sourcing notes p0 wiring ×1 every screw terminal (the converter, the switch)	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/ferrule_crimper	Haisstronica ratcheting ferrule crimper, AWG 24-8 ferrule_crimper	HX1574 Haisstronica available (2026-10-27)	1	$33.99	$33.99	1 alternate · 1 drop-in DROP-IN Haisstronica product page · $65.99 · available (the ferrule kit) checked 2026-10-27	Uses and sourcing notes tools ×1 the ferrules (once)	☐	☐
☐	buy/wire_12awg	BNTECHGO 12 AWG silicone wire, red and black (VESC leads) wire_12awg	SW12G68008F10C2 seller's no. SW12G68008F10C2 BNTECHGO	2	$14.98	$29.96	2 alternates · 2 drop-in DROP-IN BNTECHGO SW12G68008F25C2 (25 + 25 ft) · $32.98 checked 2026-10-27 DROP-IN RaceDayQuads product page · $2.99 · 12 AWG sold out on 2026-10-27; 16/18/20 AWG in stock checked 2026-10-27	Uses and sourcing notes electronics ×1 the pack, fuse, contactor and motor runs p0 wiring ×1 the two VESC branches (about 1.2 m red + black)	☐	☐
☐	buy/wire_14awg	BNTECHGO 14 AWG silicone wire, red and black (RS-04 branches) wire_14awg	14 AWG silicone red + black 25 + 25 ft BNTECHGO not shown (2026-10-27)	1	$24.98	$24.98	1 alternate · 1 drop-in DROP-IN RaceDayQuads product page · $2.99 · 14 AWG sold out on 2026-10-27 checked 2026-10-27	Uses and sourcing notes p0 wiring ×1 the four RS-04 branches, the frame bond (W33), the turret's branch and the J501's 12 V (about 4 m red + black)	☐	☐
☐	buy/wire_8awg	BNTECHGO 8 AWG silicone wire, black, 10 ft (the pack leads, the MRBF, the shunt, the switch, the bus bars; red heat-shrink on every + end) wire_8awg	8 AWG silicone black 10 ft BNTECHGO not shown (2026-10-27)	1	$17.98	$17.98	1 alternate · 1 drop-in DROP-IN BNTECHGO product page · $24.98 · not shown; 10 AWG 25 ft: carries the 80 A pulses, 55 A continuous (the MCU then caps the bus at 50 A) checked 2026-10-27	Uses and sourcing notes p0 wiring ×1 the bay's lead to the MRBF, the shunt, the switch and the bus bars (about 1.5 m; red heat-shrink on the + runs)	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/wire_22awg	BNTECHGO 22 AWG silicone wire, red and black (5 V, signals; twist the CAN pairs by hand) wire_22awg	22 AWG silicone red + black 25 + 25 ft BNTECHGO not shown (2026-10-27)	1	$11.48	$11.48	1 alternate · 1 drop-in DROP-IN RaceDayQuads product page · $1.99 · 22 AWG sold out on 2026-10-27; 20 AWG in stock checked 2026-10-27	Uses and sourcing notes p0 wiring ×1 5 V and signal wiring; the CAN pairs twisted by hand (about 6 m)	☐	☐
☐	buy/wire_18awg	BNTECHGO 18 AWG silicone wire, red and black (converters, 24 V and 12 V) wire_18awg	18 AWG silicone red + black 10 + 10 ft BNTECHGO not shown (2026-10-27)	1	$9.48	$9.48	1 alternate · 1 drop-in DROP-IN RaceDayQuads product page · $1.99 · 18 AWG in stock checked 2026-10-27	Uses and sourcing notes p0 wiring ×1 the converter's input, the 12 V loads (about 2 m red + black)	☐	☐
☐	buy/proto-board-52x50	Double-sided prototype board 50 x 70 mm, cut to size proto-board-52x50	double-sided prototype board 50 x 70 mm seller's no. PRT-08619 SparkFun In stock (2026-10-27)	5	$7.25	$36.25	3 alternates · 3 drop-in DROP-IN Tayda Electronics product page checked ? DROP-IN Tayda Electronics product page checked ? DROP-IN Tayda Electronics product page checked ?	Uses and sourcing notes turret ×1 cartridge_board turret ×1 base_board turret ×3 turret_board	☐	☐
☐	buy/sparkfun-rob-13065	Slip Ring - 12 Wire (2A) sparkfun-rob-13065	ROB-13065 SparkFun	1	$27.95	$27.95	1 alternate · 1 drop-in DROP-IN Adafruit 1196 · $19.95 · 6 checked 2026-10-27	Uses and sourcing notes turret ×1 slip_ring	☐	☐
☐	buy/hx711	HX711 load cell amplifier hx711	SEN-13879 seller's no. SEN-13879 SparkFun In stock (2026-10-27)	2	$4.95	$9.90	2 alternates · 2 drop-in DROP-IN Adafruit 5974 · $9.95 checked 2026-10-27 DROP-IN Adafruit 5974 · $9.95 checked 2026-10-27	Uses and sourcing notes electronics ×1 the load plate (stand and squat) bench ×1 B3, B4: the load cell's own amplifier on the rig node	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/resistor_kit	Resistor kit 1/4 W, 500 (incl. 2.2k, 4.7k, 10k, 100k) resistor_kit	COM-10969 seller's no. COM-10969 SparkFun	1	$9.00	$9.00	1 alternate · 1 drop-in DROP-IN Mouser 474-COM-10969 (SparkFun COM-10969) · $9.95 · 112 in stock (Mouser's figure as listed on findchips.com today) checked 2026-10-27	Uses and sourcing notes electronics ×1 the node's dividers: the pendant loop 10k/2.2k (12 V -> 2.2 V), the motors' bus 100k/4.7k (60 V -> 2.7 V)	☐	☐
☐	buy/proto-board-70x50	Double Side Prototyping Board 50x70mm proto-board-70x50	double-sided prototype board 50 x 70 mm seller's no. PRT-08619 SparkFun In stock (2026-10-27)	1	$7.25	$7.25	See note below	See note below	☐	☐
Alternates: 3 alternates · 1 drop-in CHANGES Botland Universal double sided board 50x70mm (Botland PLY-02745, EAN 5904422303303) · EUR 0.90 · Orderable Same board class as the primary. Mounting-hole positions are not given, so the stand's bosses may need to move (see notes). checked 2026-10-27 CHANGES Adafruit Adafruit 866 'Snap-action 3-Wire Block Connector (12-24 AWG) - Pack of 3' / Adafruit 874 '5-Wire' (brand not confirmed as WAGO 221) No board: lever connectors splice the 24 V feed (XT60 -> fuse -> switch -> DT12 plug power pins) and the pendant loop in free air. Drop the four M3 inserts and bosses for the 70 x 50 board (or keep them for a strain-relief plate), cable-tie the connectors to the floor or a printed cradle, and use 12-24 AWG wire. There is no screw-terminal board to service, and the pendant loop becomes two inline splices. checked 2026-10-27 DROP-IN Tayda Electronics product page checked ? **Part details:** Uses and sourcing notes stand ×1 bench DOESN'T FIT AS DRAWN The stand's four M3 bosses sit on 66 x 50 mm centres, at the edges of a 70 x 50 board; its holes are a few mm in from the edges (probably 44-46 mm apart across). Measure the board and move the bosses. (2026-10-27)									
☐	buy/diode_1n4001	Rectifier diode 1N4001 (1 A, 50 V) diode_1n4001	COM-08589 seller's no. COM-08589 SparkFun In stock (2026-10-27)	2	$0.250	$0.50	1 alternate · 1 drop-in DROP-IN Adafruit 755 (1N4001 Diode - 10 pack) · $1.50 / 10 · 38 in stock checked 2026-10-27	Uses and sourcing notes electronics ×2 across the contactor's coil (cathode to +): spares the relay's contact the coil's kick; one spare	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/bench_supply_36v	36 V 16.6 A enclosed supply bench_supply_36v	Mean Well LRS-600-36 seller's no. Mean Well LRS-600-36 Bravo Electro	1	$61.60	$61.60	1 alternate · 1 drop-in DROP-IN TRC Electronics LRS-600-36 · $66.80 · Contact Us (lead time from 1 week) checked 2026-10-27	Uses and sourcing notes electronics ×1 bench tests only (B2, the first enable, B1's tested motor): the robot's 36 V; it cannot sink regeneration	☐	☐
☐	buy/dcdc_ddr30l12	DIN-rail DC-DC, 18-75 V in, 12 V 2.5 A out dcdc_ddr30l12	Mean Well DDR-30L-12 seller's no. Mean Well DDR-30L-12 Bravo Electro In stock (2026-10-27)	1	$27.60	$27.60	1 alternate · 1 drop-in DROP-IN TRC Electronics DDR-30L-12 · $35.36 · 10 can ship now checked 2026-10-27	Uses and sourcing notes electronics ×1 the 12 V rail: the contactor coil, the timer, the pendant loop	☐	☐
☐	buy/switchcraft-ta3flx	Mini-XLR 3-pin female cable plug (TA3FLX), latching, cable <= 4.32 mm switchcraft-ta3flx	TA3FLX seller's no. 502-TA3FLX Mouser 323 (2026-10-27)	1	$14.22	$14.22	1 alternate · 1 drop-in DROP-IN Digi-Key TA3FLX-ND · $14.22 · 631 checked 2026-10-27	Uses and sourcing notes stand ×1 bench	☐	☐
☐	buy/switchcraft-tb3m	Mini-XLR 3-pin male panel receptacle (TB3M), latching, keyed switchcraft-tb3m	TB3M seller's no. 502-TB3M Mouser 2619 (2026-10-27)	1	$13.10	$13.10	1 alternate · 1 drop-in DROP-IN Digi-Key TB3M-ND · $13.24 · 1488 checked 2026-10-27	Uses and sourcing notes stand ×1 bench	☐	☐
☐	buy/littelfuse-0fhm0001zxj	In-line holder for a MINI blade fuse, black leads (MINI FHM series) littelfuse-0fhm0001zxj	0FHM0001ZXJ seller's no. 576-0FHM0001ZXJ Mouser 8130 (2026-10-27)	1	$9.47	$9.47	1 alternate · 1 drop-in DROP-IN Digi-Key product page checked 2026-10-27	Uses and sourcing notes turret ×1 el_fuse_holder	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/te-dt04-12pa-l012	DT 12-way receptacle (pin side), key A, grey, flange mount (L012 mod) - LCSC C6423585 te-dt04-12pa-l012	DT04-12PA-L012 seller's no. 571-DT04-12PA-L012 Mouser 7037 (2026-10-27)	1	$7.38	$7.38	See note below	See note below	☐	☐
Alternates: 5 alternates · 4 drop-in DROP-IN Newark DT04-12PA-L012 (Newark 38AH5620) · about $14.32 to $16.02 depending on quantity (search snippet); an oemstrade aggregator snippet showed Newark at $15.36 (1) / $14.91 (10) · In stock (99) checked 2026-10-27 DROP-IN Del City DT04-12PA-L012 (Del City 15061229) · $12.84 at 1+ (search snippet) · In stock checked 2026-10-27 DROP-IN Chief Enterprises AT04-12PA-L012 (Amphenol Sine Systems) · $3.75 (250-999), $3.46 (1000-9999); minimum purchase 250 units · Out of stock; 133 days lead time checked 2026-10-27 CHANGES Waytek AT04-12PA-PM11 (Amphenol Sine Systems; Waytek #38157) Different flange. Amphenol drawing AT04-12PX-PMXX shows a one-piece flange with gasket and 2 holes (2X dia 4.0, 16.4 deep) for #10 sheet-metal or ST 4.8 plastic screws (mounting plate dims 45.50 / 61.0 / 25.50 mm), not the 4-hole L012 flange. The turret-base cutout and screw holes would change. It also needs the AW12P wedge. checked 2026-10-27 DROP-IN LCSC C6423585 · $10.36 checked 2026-10-27 **Part details:** Uses and sourcing notes turret ×1 dt_receptacle PARTS MISSING The receptacle also needs a W12P wedge lock and 12 size-16 pin contacts (TE: sold separately), which are not on the list. Buy plug and receptacle from one family (TE DT with TE DT, or Amphenol AT with AT). (2026-10-27)									
☐	buy/molex-43030-0010	Micro-Fit 3.0 female crimp terminal, tin, 26-30 AWG molex-43030-0010	43030-0010 seller's no. 538-43030-0010 Mouser 287656 (2026-10-27)	30	$0.200	$6.00	1 alternate · 1 drop-in DROP-IN Digi-Key WM1838-ND checked 2026-10-27	Uses and sourcing notes turret ×30 el_microfit_f28	☐	☐
☐	buy/cap-mlcc-100n-50v	Capacitor 100 nF 50 V X7R radial ceramic, 2.5 mm leads cap-mlcc-100n-50v	K104K15X7RF53L2 seller's no. 594-K104K15X7RF53L2 Mouser 57498 (2026-10-27)	18	$0.250	$4.50	1 alternate · 1 drop-in DROP-IN Digi-Key BC3324-ND · $0.29 checked 2026-10-27	Uses and sourcing notes turret ×18 el_c_100n	☐	☐
☐	buy/tvs-p6ke15a	TVS diode 600 W unidirectional, 12.8 V standoff, 21.2 V clamp, DO-15 (P6KE15A) tvs-p6ke15a	P6KE15A-E3/54 seller's no. 625-P6KE15A-E3 Mouser 2378 (2026-10-27)	3	$0.960	$2.88	1 alternate · 1 drop-in DROP-IN Digi-Key P6KE15A-E3/54GICT-ND · $0.96 · 1054 checked 2026-10-27	Uses and sourcing notes turret ×3 el_d_tvs15	☐	☐
☐	buy/cap-mlcc-1u-50v	Capacitor 1 uF 50 V X7R radial ceramic, 5 mm leads cap-mlcc-1u-50v	FG24X7R1H105KRT06 seller's no. 810-FG24X7R1H105KRT6 Mouser 11666 (2026-10-27)	4	$0.540	$2.16	1 alternate · 1 drop-in DROP-IN Digi-Key 445-173374-1-ND · $0.51 checked 2026-10-27	Uses and sourcing notes turret ×4 el_c_1u	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/cap-film-100n-5pct	Capacitor 100 nF 63 V +-5 % PET film, 5 mm leads cap-film-100n-5pct	MKS2C031001A00JSSD seller's no. 505-MKS2.1/63/5 Mouser 2166 (2026-10-27)	3	$0.700	$2.10	1 alternate · 1 drop-in DROP-IN Newark 46T6775 · $0.45 · 9338 checked 2026-10-27	Uses and sourcing notes turret ×3 el_c_100n_film	☐	☐
☐	buy/schottky-1n5822	Schottky rectifier 3 A 40 V, DO-201AD (1N5822) schottky-1n5822	1N5822G seller's no. 863-1N5822G Mouser 15695 (2026-10-27)	2	$0.830	$1.66	1 alternate · 1 drop-in DROP-IN Digi-Key 1N5822-E3/54GICT-ND · $0.91 · 7531 checked 2026-10-27	Uses and sourcing notes turret ×2 el_d_1n5822	☐	☐
☐	buy/yageo-mfr-25fbf52-100k	Resistor 100 k 1 % 1/4 W metal film (Yageo MFR-25FBF52) yageo-mfr-25fbf52-100k	MFR-25FBF52-100K seller's no. 603-MFR-25FBF52-100K Mouser 15007 (2026-10-27)	16	$0.100	$1.60	1 alternate · 1 drop-in DROP-IN Digi-Key 100KXBK-ND checked 2026-10-27	Uses and sourcing notes turret ×16 el_r_100k	☐	☐
☐	buy/yageo-mfr-25fbf52-1k	Resistor 1.00 k 1 % 1/4 W metal film (Yageo MFR-25FBF52) yageo-mfr-25fbf52-1k	MFR-25FBF52-1K seller's no. 603-MFR-25FBF52-1K Mouser 51348 (2026-10-27)	16	$0.100	$1.60	1 alternate · 1 drop-in DROP-IN Digi-Key 1.00KXBK-ND · $0.10 checked 2026-10-27	Uses and sourcing notes turret ×16 el_r_1k	☐	☐
☐	buy/yageo-mfr-25fbf52-10k	Resistor 10 k 1 % 1/4 W metal film (Yageo MFR-25FBF52) yageo-mfr-25fbf52-10k	MFR-25FBF52-10K seller's no. 603-MFR-25FBF52-10K Mouser 114923 (2026-10-27)	14	$0.100	$1.40	1 alternate · 1 drop-in DROP-IN Digi-Key 10.0KXBK-ND · $0.10 checked 2026-10-27	Uses and sourcing notes turret ×14 el_r_10k	☐	☐
☐	buy/ti-cd74hc423e	Dual retriggerable monostable, 74HC423 (no trigger on a rising reset), PDIP-16 ti-cd74hc423e	CD74HC423E seller's no. 595-CD74HC423E Mouser 453 (2026-10-27)	1	$1.25	$1.25	1 alternate · 1 drop-in DROP-IN Digi-Key 296-9229-5-ND · $1.22 · 296 checked 2026-10-27	Uses and sourcing notes turret ×1 el_monostables	☐	☐
☐	buy/diode-1n5408g	Rectifier 3 A 1000 V, DO-201AD (1N5408) diode-1n5408g	1N5408G seller's no. 863-1N5408G Mouser 33554 (2026-10-27)	2	$0.580	$1.16	1 alternate · 1 drop-in DROP-IN Newark 42K2941 · $0.62 · 31895 checked 2026-10-27	Uses and sourcing notes turret ×2 el_d_1n5408	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/vishay-bat85	Schottky diode 30 V 200 mA, DO-35 (BAT85S) vishay-bat85	BAT85S-TAP seller's no. 78-BAT85S-TAP Mouser 41995 (2026-10-27)	2	$0.500	$1.00	1 alternate · 1 drop-in DROP-IN Digi-Key BAT85S-TAPCT-ND · $0.50 · 61329 checked 2026-10-27	Uses and sourcing notes turret ×2 el_d_bat85	☐	☐
☐	buy/ti-sn74hct14n	Hex Schmitt-trigger inverter SN74HCT14N (DIP-14) ti-sn74hct14n	SN74HCT14N seller's no. 595-SN74HCT14N Mouser 3467 (2026-10-27)	1	$0.920	$0.92	1 alternate · 1 drop-in DROP-IN Digi-Key 296-8394-5-ND · $0.87 checked 2026-10-27	Uses and sourcing notes turret ×1 el_brake_driver	☐	☐
☐	buy/schottky-1n5817	Schottky rectifier 1 A 20 V, DO-41 (1N5817) schottky-1n5817	1N5817-T seller's no. 621-1N5817 Mouser 15381 (2026-10-27)	2	$0.370	$0.74	1 alternate · 1 drop-in DROP-IN Digi-Key 1N5817DICT-ND · $0.37 · 28734 checked 2026-10-27	Uses and sourcing notes turret ×2 el_d_1n5817	☐	☐
☐	buy/omron-b3f-1000	Tactile switch 6x6 mm, 4.3 mm, 100 gf (B3F-1000) omron-b3f-1000	B3F-1000 seller's no. 653-B3F-1000 Mouser 25871 (2026-10-27)	2	$0.350	$0.70	1 alternate · 1 drop-in DROP-IN Digi-Key SW400-ND · $0.35 checked 2026-10-27	Uses and sourcing notes turret ×2 el_sw_reset	☐	☐
☐	buy/bourns-mf-r010	Resettable PTC fuse 0.10 A hold, 60 V, radial (MF-R010) bourns-mf-r010	MF-R010 seller's no. 652-MFR010 Mouser 11219 (2026-10-27)	2	$0.330	$0.66	1 alternate · 1 drop-in DROP-IN Digi-Key MF-R010-ND · $0.40 · 7433 checked 2026-10-27	Uses and sourcing notes turret ×2 el_polyfuse_loop	☐	☐
☐	buy/microchip-mcp120	Voltage supervisor 2.93 V, open-drain active-low reset, TO-92 microchip-mcp120	MCP120-300DI/TO seller's no. 579-MCP120-300DI/TO Mouser 894 (2026-10-27)	1	$0.600	$0.60	1 alternate · 1 drop-in DROP-IN Digi-Key MCP120-300DI/TO-ND · $0.60 · 1901 checked 2026-10-27	Uses and sourcing notes turret ×1 el_supervisor	☐	☐
☐	buy/yageo-rsf100jb-73-2k2	Resistor 2.2k 1 W 5 % metal oxide (RSF100) yageo-rsf100jb-73-2k2	RSF100JB-73-2K2 seller's no. 603-RSF100JB-73-2K2 Mouser 9082 (2026-10-27)	3	$0.200	$0.60	1 alternate · 1 drop-in DROP-IN Digi-Key 2.2KW-1-ND checked 2026-10-27	Uses and sourcing notes turret ×3 el_r_2k2_1w	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/yageo-mfr-25fbf52-39k	Resistor 39 k 1 % 1/4 W metal film (Yageo MFR-25FBF52) yageo-mfr-25fbf52-39k	MFR-25FBF52-39K seller's no. 603-MFR-25FBF52-39K Mouser 5471 (2026-10-27)	5	$0.100	$0.50	1 alternate · 1 drop-in DROP-IN Digi-Key 39.0KXBK-ND checked 2026-10-27	Uses and sourcing notes turret ×5 el_r_39k	☐	☐
☐	buy/onsemi-1n4148	Diode 1N4148 (DO-35) onsemi-1n4148	1N4148 seller's no. 512-1N4148 Mouser 1090369 (2026-10-27)	4	$0.100	$0.40	1 alternate · 1 drop-in DROP-IN Digi-Key 1N4148FS-ND · $0.10 checked 2026-10-27	Uses and sourcing notes turret ×4 el_d_1n4148	☐	☐
☐	buy/yageo-mfr-25fbf52-100r	Resistor 100 R 1 % 1/4 W metal film (Yageo MFR-25FBF52) yageo-mfr-25fbf52-100r	MFR-25FBF52-100R seller's no. 603-MFR-25FBF52-100R Mouser 49654 (2026-10-27)	4	$0.100	$0.40	1 alternate · 1 drop-in DROP-IN Digi-Key 100XBK-ND checked 2026-10-27	Uses and sourcing notes turret ×4 el_r_100r	☐	☐
☐	buy/yageo-mfr-25fbf52-6k8	Resistor 6.8 k 1 % 1/4 W metal film (Yageo MFR-25FBF52) yageo-mfr-25fbf52-6k8	MFR-25FBF52-6K8 seller's no. 603-MFR-25FBF52-6K8 Mouser 12179 (2026-10-27)	3	$0.100	$0.30	1 alternate · 1 drop-in DROP-IN Digi-Key 13-MFR-25FRF52-6K8CT-ND · $0.10 checked 2026-10-27	Uses and sourcing notes turret ×3 el_r_6k8	☐	☐
☐	buy/yageo-mfr-25fbf52-4k7	Resistor 4.7 k 1 % 1/4 W metal film (Yageo MFR-25FBF52) yageo-mfr-25fbf52-4k7	MFR-25FBF52-4K7 seller's no. 603-MFR-25FBF52-4K7 Mouser 12937 (2026-10-27)	2	$0.110	$0.22	1 alternate · 1 drop-in DROP-IN Digi-Key product page checked 2026-10-27	Uses and sourcing notes turret ×2 el_r_4k7	☐	☐
☐	buy/yageo-mfr-25fbf52-162r	Resistor 162 R 1 % 1/4 W metal film (Yageo MFR-25FBF52) yageo-mfr-25fbf52-162r	MFR-25FBF52-162R seller's no. 603-MFR-25FBF52-162R Mouser 7377 (2026-10-27)	2	$0.100	$0.20	1 alternate · 1 drop-in DROP-IN Arrow V99:2348_17733694 · $0.02 · 8185 (Americas) checked 2026-10-27	Uses and sourcing notes turret ×2 el_r_162r	☐	☐
☐	buy/yageo-mfr-25fbf52-182k	Resistor 182 k 1 % 1/4 W metal film (Yageo MFR-25FBF52) yageo-mfr-25fbf52-182k	MFR-25FBF52-182K seller's no. 603-MFR-25FBF52-182K Mouser 31912 (2026-10-27)	2	$0.100	$0.20	1 alternate · 1 drop-in DROP-IN Quest Components product page · $0.05 · 440 checked 2026-10-27	Uses and sourcing notes turret ×2 el_r_182k	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/yageo-mfr-25fbf52-1k33	Resistor 1.33 k 1 % 1/4 W metal film (Yageo MFR-25FBF52) yageo-mfr-25fbf52-1k33	MFR-25FBF52-1K33 seller's no. 603-MFR-25FBF52-1K33 Mouser 17999 (2026-10-27)	2	$0.100	$0.20	1 alternate · 1 drop-in DROP-IN Arrow V99:2348_17668792 · $0.02 · 3495 (Americas) checked 2026-10-27	Uses and sourcing notes turret ×2 el_r_1k33	☐	☐
☐	buy/yageo-mfr-25fbf52-20r	Resistor 20.0 R 1 % 1/4 W metal film (Yageo MFR-25FBF52) yageo-mfr-25fbf52-20r	MFR-25FBF52-20R seller's no. 603-MFR-25FBF52-20R Mouser 3236 (2026-10-27)	2	$0.100	$0.20	1 alternate · 1 drop-in DROP-IN Arrow V99:2348_17665225 · $0.02 · 6692 checked 2026-10-27	Uses and sourcing notes turret ×2 el_r_20r	☐	☐
☐	buy/yageo-mfr-25fbf52-324r	Resistor 324 R 1 % 1/4 W metal film (Yageo MFR-25FBF52) yageo-mfr-25fbf52-324r	MFR-25FBF52-324R seller's no. 603-MFR-25FBF52-324R Mouser 7829 (2026-10-27)	2	$0.100	$0.20	1 alternate · 1 drop-in DROP-IN Arrow V99:2348_17662115 · $0.02 · 9514 checked 2026-10-27	Uses and sourcing notes turret ×2 el_r_324r	☐	☐
☐	buy/yageo-mfr-25fbf52-40r2	Resistor 40.2 R 1 % 1/4 W metal film (Yageo MFR-25FBF52) yageo-mfr-25fbf52-40r2	MFR-25FBF52-40R2 seller's no. 603-MFR-25FBF52-40R2 Mouser 5209 (2026-10-27)	2	$0.100	$0.20	1 alternate · 1 drop-in DROP-IN Quest Components product page · $0.06 · 557 checked 2026-10-27	Uses and sourcing notes turret ×2 el_r_40r2	☐	☐
☐	buy/yageo-mfr-25fbf52-60k4	Resistor 60.4 k 1 % 1/4 W metal film (Yageo MFR-25FBF52) yageo-mfr-25fbf52-60k4	MFR-25FBF52-60K4 seller's no. 603-MFR-25FBF52-60K4 Mouser 12929 (2026-10-27)	2	$0.100	$0.20	1 alternate · 1 drop-in DROP-IN Digi-Key 60.4KXBK-ND checked 2026-10-27	Uses and sourcing notes turret ×2 el_r_60k4	☐	☐
☐	buy/yageo-mfr-25fbf52-63k4	Resistor 63.4 k 1 % 1/4 W metal film (Yageo MFR-25FBF52) yageo-mfr-25fbf52-63k4	MFR-25FBF52-63K4 seller's no. 603-MFR-25FBF52-63K4 Mouser 8863 (2026-10-27)	2	$0.100	$0.20	1 alternate · 1 drop-in DROP-IN Digi-Key product page checked 2026-10-27	Uses and sourcing notes turret ×2 el_r_63k4	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/yageo-mfr-25fbf52-66k5	Resistor 66.5 k 1 % 1/4 W metal film (Yageo MFR-25FBF52) yageo-mfr-25fbf52-66k5	MFR-25FBF52-66K5 seller's no. 603-MFR-25FBF52-66K5 Mouser 19592 (2026-10-27)	2	$0.100	$0.20	1 alternate · 1 drop-in DROP-IN Digi-Key 66.5KXBK-ND checked 2026-10-27	Uses and sourcing notes turret ×2 el_r_66k5	☐	☐
☐	buy/yageo-mfr-25fbf52-69k8	Resistor 69.8 k 1 % 1/4 W metal film (Yageo MFR-25FBF52) yageo-mfr-25fbf52-69k8	MFR-25FBF52-69K8 seller's no. 603-MFR-25FBF52-69K8 Mouser 14226 (2026-10-27)	2	$0.100	$0.20	1 alternate · 1 drop-in DROP-IN Digi-Key 13-MFR-25FBF52-69K8-ND checked 2026-10-27	Uses and sourcing notes turret ×2 el_r_69k8	☐	☐
☐	buy/yageo-mfr-25fbf52-73k2	Resistor 73.2 k 1 % 1/4 W metal film (Yageo MFR-25FBF52) yageo-mfr-25fbf52-73k2	MFR-25FBF52-73K2 seller's no. 603-MFR-25FBF52-73K2 Mouser 8644 (2026-10-27)	2	$0.100	$0.20	1 alternate · 1 drop-in DROP-IN Digi-Key 13-MFR-25FBF52-73K2-ND checked 2026-10-27	Uses and sourcing notes turret ×2 el_r_73k2	☐	☐
☐	buy/yageo-mfr-25fbf52-80r6	Resistor 80.6 R 1 % 1/4 W metal film (Yageo MFR-25FBF52) yageo-mfr-25fbf52-80r6	MFR-25FBF52-80R6 seller's no. 603-MFR-25FBF52-80R6 Mouser 18577 (2026-10-27)	2	$0.100	$0.20	1 alternate · 1 drop-in DROP-IN Digi-Key product page checked 2026-10-27	Uses and sourcing notes turret ×2 el_r_80r6	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/bumper-feet-12.7	3M Bumpon SJ5003 - hemisphere, 0.440 in (11.2 mm) diameter x 0.200 in (5.1 mm) high bumper-feet-12.7	SJ5003 (Bumpon) seller's no. 517-SJ-5003BK Mouser 187488 (2026-10-27)	4 buy 1 pack of 56	$0.0023 $0.13 per 56	$0.13 used $0.01	See note below	Uses and sourcing notes stand ×4 bench TO CHECK No 3M Bumpon is 12.7 x 5.5 mm. The SJ5003 (11.2 x 5.1 mm dome) sits in the 13.2 mm recess with 1 mm play and stands 4.1 mm proud instead of 4.5. (2026-10-27)	☐	☐
Alternates: 3 alternates · 1 drop-in CHANGES CabinetParts.com 3M Bumpon SJ5312 (clear cylinder, 12.7 x 3.6 mm) Correct 12.7 mm diameter (sits centred in the 13.2 mm recess) but only 3.6 mm tall, so it stands about 2.6 mm proud instead of 4.5. Update the foot envelope height in cad/turret_stand.py. checked 2026-10-27 CHANGES Adafruit Adafruit 550 'Little Rubber Bumper Feet - Pack of 4' · $0.95 per pack of 4 (1-9 packs), $0.86 (10-99), $0.76 (100+) · In stock Only about 8 mm diameter x 4 mm tall (style varies by supplier). It would sit loose in the 13.2 mm recess and stand about 3 mm proud. Shrink the recesses to about 8.5 mm or accept off-centre feet. checked 2026-10-27 DROP-IN Digi-Key product page checked ?										
☐	buy/timer_offdelay	Timer relay, multifunction (off-delay with a trigger input), 0.1-10 s, 12-125 VDC / 240 VAC, 10 A timer_offdelay	ProSense T2R-M1-ADJ-240U seller's no. ProSense T2R-M1-ADJ-240U AutomationDirect 76 Available (2026-10-27)	1	$44.50	$44.50	2 alternates · 2 drop-in DROP-IN DROK 200350 (LCD multifunction delay relay module, 6-30 V) · $6.55 checked 2026-10-27 DROP-IN Galco Finder 80.01.0.240.0000 · $78.85 · ships 11/6 checked 2026-10-27	Uses and sourcing notes electronics ×1 holds the contactor's coil 1.0 s after the pendant is let go	☐	☐
☐	buy/can_cable	Shielded 22 AWG 1.5-pair data cable, 20 ft can_cable	Quabbin Q0357-1 seller's no. Quabbin Q0357-1 AutomationDirect	1	$21.20	$21.20	1 alternate · 1 drop-in DROP-IN Pro Wire & Cable Belden 3106A · $82.60 · 823136 in stock; Typically Ships in 1-2 Days checked 2026-10-27	Uses and sourcing notes electronics ×1 the CAN chain (pair + ground) and the 12 V signal runs (pendant, timer, coil: under 0.5 A)	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/main_switch	Flipsky Anti Spark Switch Smart Enhanced 300A V2.0, 12-84 V (3-20S), 100 A continuous, 16 mm button main_switch	ANSIFS20SH Flipsky ships from CN available (2026-10-27)	1	$52.00	$52.00	1 alternate · 1 drop-in DROP-IN Flipsky product page · $48.00 · available; Antispark Pro V3.0 (AWAPFS14SH), 12-60 V: 53.3 V full is inside it checked 2026-10-27	Uses and sourcing notes p0 power skid ×1 the main switch: low side, between the bay's - and the - bus; momentary mode (D54)	☐	☐
☐	buy/load_cell_s500	S-type load cell 500 kg (C2), M12 x 1.75 both ends, 51 x 76 x 19 mm load_cell_s500	3140_0 seller's no. 3140_0 Phidgets Quantity Available: 358 (2026-10-27)	1	$50.00	$50.00	2 alternates · 2 drop-in DROP-IN RobotShop Phidgets S-Type Load Cell 500 kg (C3) · $71.54 checked 2026-10-27 DROP-IN SparkFun SEN-14282 (TAS501, 200 kg) · $96.95 · In stock checked 2026-10-27	Uses and sourcing notes bench ×1 the press cage (B3 the stop sleeve to 2 kN, B4 the inserts and the link screws to 3.5 kN), read by an HX711	☐	☐
☐	buy/sunnysky-x2216-1400kv-v3	SunnySky X2216 1400KV V3 outrunner, short shaft, with its bolt-on prop adapter and X-mount sunnysky-x2216-1400kv-v3	X2216 V3 1400KV, short shaft (SKU SS-XV3-X2216-S-1400-V3) seller's no. SS-XV3-X2216-S-1400-V3 (variant 27868073427038, '1400 Short') SunnySky USA available: true (Shopify; no count exposed) (2026-10-27)	2	$24.99	$49.98	See note below	Uses and sourcing notes turret ×1 flywheel_motor_l turret ×1 flywheel_motor_r	☐	☐
Alternates: 2 alternates · 2 drop-in DROP-IN ReadyMadeRC SS-XV3-X2216-S-1400-V3 (RMRC 84546, GTIN 680306447800) · $24.99 · In Stock (no quantity shown) checked 2026-10-27 DROP-IN ReadyMadeRC SS-XV3-X2216-S-1400-V3 (RMRC 84546, GTIN 680306447800) · $24.99 · In Stock (no quantity shown) checked 2026-10-27										

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/dial_indicator	Dial indicator 0-1 in x 0.001 in with a magnetic base dial_indicator	G9849 (Grizzly Magnetic Base/Dial Indicator Combo) seller's no. G9849 (Grizzly Magnetic Base/Dial Indicator Combo) Grizzly Available; Delivery est. by 11/2 - 11/5; Available in Bellingham, WA and Springfield, MO (2026-10-27)	1	$49.95	$49.95	1 alternate · 1 drop-in DROP-IN Shars 303-2724 · $66.95 · In stock checked 2026-10-27	Uses and sourcing notes bench ×1 B3 and B5: the sleeve's and the tyre's deflection	☐	☐
☐	buy/blue-sea-6006	m-Series Mini ON-OFF battery switch, single circuit, 48 V DC, 275-300 A continuous (D70: the stand's DC power switch, 15 A or more) blue-sea-6006	6006 Walmart	1	$34.16	$34.16	2 alternates · 2 drop-in DROP-IN NavStore product page · $27.95 checked 2026-10-28 DROP-IN Waytek product page · $39.96 checked 2026-10-28	Uses and sourcing notes stand ×1 bench	☐	☐
☐	buy/dewalt-20v-dock-adapter-12awg	DeWalt 20 V MAX battery dock adapter, 12 AWG leads (takes a FLEXVOLT pack in its 20 V mode) dewalt-20v-dock-adapter-12awg	DeWalt 18/20 V MAX battery dock adapter, 12 AWG leads Walmart	1	$12.95	$12.95	1 alternate · 1 drop-in DROP-IN Etsy product page · $13.50 checked ?	Uses and sourcing notes stand ×1 bench	☐	☐
☐	buy/tool-deutsch-crimper-iwd1620	IWISS IWD-1620 crimper for Deutsch DT size-16 solid contacts, 16-20 AWG (tool) tool-deutsch-crimper-iwd1620	IWD-1620 (Shopify variant 40373778415729) seller's no. IWD-1620 (Shopify variant 40373778415729) IWISS available: true (2026-10-27)	1	$24.99	$24.99	1 alternate · 1 drop-in DROP-IN McMaster-Carr 6084T3 · $64.98 · 'Delivers tomorrow 7-9 am' checked 2026-10-27	Uses and sourcing notes tools ×1 tool	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/ferrule_kit	Ferrule crimper + 770 ferrules (23-10 AWG) ferrule_kit	HSC8 6-4-KIT seller's no. HSC8 6-4-KIT IWISS	1	$19.99	$19.99	1 alternate · 1 drop-in DROP-IN Amazon (sold by iwiss, ships from Amazon) ASIN B07LCF39W9 · $24.99 · In Stock checked 2026-10-27	Uses and sourcing notes electronics ×1 every screw terminal (fuse holder, contactor coil, timer, DC-DC)	☐	☐
☐	buy/worker-cheetah-rival-rounds-100	Worker Cheetah Rival-compatible rounds, 100-pack worker-cheetah-rival-rounds-100	Cheetah Rival-compatible rounds, 100-pack Out of Darts	2	$14.99	$29.98	2 alternates · 2 drop-in DROP-IN Amazon ASIN B071SC9QYG · $14.99 · orderable (Add to cart shown); Amazon's Choice 'Overall Pick', 4.7 stars (8.9K), '400+ bought in past month' checked 2026-10-27 DROP-IN Out of Darts product page checked ?	Uses and sourcing notes stand ×2 ammo	☐	☐
☐	buy/hasbro-nerf-rival-accu-round-60	Nerf Rival 60 Accu-Round Refill (Hasbro F5052), 60 official Rival rounds — Worker Cheetah 100-pack (Out of Darts) hasbro-nerf-rival-accu-round-60	Worker Cheetah 100-pack Out of Darts In stock (2026-10-27)	1	$14.99	$14.99	3 alternates · 3 drop-in DROP-IN Walmart (the catalogue's) F5052 · $19.99 each checked 2026-10-27 DROP-IN Walmart Nerf Rival 50-Round Refill · $15.95 50 rounds, not 60 checked 2026-10-27 DROP-IN eBay product page · $16.95 checked ?	Uses and sourcing notes stand ×1 ammo the plan's seller (docs/turret-prototype-plan.md 3.3, Primary) instead of the catalogue's Hasbro (Nerf) F5052	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/fuse_kldr030	Class CC time-delay fuse 30 A, 600 VAC / 300 VDC fuse_kldr030	Littelfuse KLDR030.TXP seller's no. Littelfuse KLDR030.TXP Standard Electric Supply 812 In Stock (2026-10-27)	1	$31.70	$31.70	2 alternates · 2 drop-in DROP-IN Galco Littelfuse KLDR30 · $44.33 · 10 In Stock checked 2026-10-27 DROP-IN Galco Mersen ATDR30 · $19.96 · 31 In Stock checked 2026-10-27	Uses and sourcing notes electronics ×1 the pack's positive, before the contactor	☐	☐
☐	buy/lug_8awg_m8	Selterm tinned copper lug 8 AWG, 5/16 in (M8) stud, 10 pcs (the MRBF block) lug_8awg_m8	MD0856P Selterm available (2026-10-27)	4 buy 1 pack of 10	$1.61 $16.15 per 10	$16.15 used $6.46	1 alternate · 1 drop-in DROP-IN go2marine product page · $4.95 / 3 · available; Ancor 12-10 AWG M8 checked 2026-10-27	Uses and sourcing notes p0 wiring ×4 the MRBF block's M8 studs and the shunt	☐	☐
☐	buy/lug_8awg_m6	Selterm tinned copper lug 8 AWG, 1/4 in (M6) stud, 10 pcs (crimped on 10 AWG with a sleeve) lug_8awg_m6	MD0814P Selterm available (2026-10-27)	10 buy 1 pack of 10	$1.50 $15.01 per 10	$15.01	1 alternate · 1 drop-in DROP-IN go2marine product page · $4.95 / 3 · available; Ancor adhesive heat-shrink rings 12-10 AWG M6 checked 2026-10-27	Uses and sourcing notes p0 wiring ×10 the bus bars' M6 taps	☐	☐
☐	buy/stud_m12x80	Fully threaded stud M12 x 75 stud_m12x80	91217237 seller's no. 91217237 Fastenal Local DC Inventory; Direct Ship: Ships TOMORROW (2026-10-27)	1	$31.02	$31.02	1 alternate · 1 drop-in DROP-IN BelMetric S12X150FCLZ · $6.88 · in stock checked 2026-10-27	Uses and sourcing notes bench ×1 the press cage: the puller hung from the load cell (B4); 75 mm reaches 10 of the nut's 11 mm	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/fuse_holder_mini_inline	Bussmann HHM inline MINI fuse holder, 2-30 A (rated 32 V DC: the 58 V fuse does the interrupting; label it "58 V fuses only") fuse_holder_mini_inline	HHM RockAuto Add to Cart (2026-10-27)	13	$2.28	$29.64	2 alternates · 2 drop-in DROP-IN LCSC product page · $0.30 · 0 (Keystone 3570 PCB holder: the fuse-board upgrade) checked 2026-10-27 DROP-IN LCSC product page · $0.30 · 0 (Keystone 3570 PCB holder: the fuse-board upgrade) checked 2026-10-27	Uses and sourcing notes stand ×1 bench p0 power skid ×12 one per branch (11) and a spare: labelled "58 V fuses only"; pull only with the packs out	☐	☐
☐	buy/fuse_holder_cc	Class CC fuse holder, 1-pole, DIN rail, 600 VAC/DC fuse_holder_cc	Littelfuse LPSC0001Z seller's no. Littelfuse LPSC0001Z Galco 170 In Stock (2026-10-27)	1	$28.99	$28.99	2 alternates · 2 drop-in DROP-IN Standard Electric Supply Littelfuse LFPSC0001Z · $26.48 · 62 In Stock checked 2026-10-27 DROP-IN Galco Mersen USCC1 · $23.18 · 68 In Stock checked 2026-10-27	Uses and sourcing notes electronics ×1 holds the 30 A fuse on the pod's DIN rail	☐	☐
☐	buy/usb_can_u2c	USB to CAN adapter (candleLight gs_usb: SocketCAN) usb_can_u2c	BIGTREETECH U2C V2.1 (SKU 1030000107) seller's no. BIGTREETECH U2C V2.1 (SKU 1030000107) BIQU Equipment ships from CN orderable (2026-10-27)	1	$23.99	$23.99	2 alternates · 2 drop-in DROP-IN InnoMaker USB2CAN-Module (isolated) · $34.99 checked 2026-10-27 DROP-IN West3D FYSETC UCAN · $19.99 checked 2026-10-27	Uses and sourcing notes electronics ×1 the laptop on the CAN bus at 1 Mbit/s	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/plywood_18	Sanded plywood 23/32 in, 2 x 4 ft project panel plywood_18	Internet #202093836 seller's no. Internet #202093836 The Home Depot	1	$23.76	$23.76	1 alternate · 1 drop-in DROP-IN Lowe's Item #7710, Model #12611 · $27.50 checked 2026-10-27	Uses and sourcing notes rig ×1 the deck (600 x 400) and the load plate's pad (400 x 300)	☐	☐
☐	buy/weight_plate_2p5	Cast iron standard plate, 2.5 lb, 1 in hole (129.5 x 20.3 mm) weight_plate_2p5	RPB2-5 seller's no. RPB2-5 Body-Solid	2	$6.65	$13.30	1 alternate · 1 drop-in DROP-IN Amazon (CAP Barbell) ASIN B0FNPPH748 (4-pack) · $17.80 / 4 checked 2026-10-27	Uses and sourcing notes rig ×2 ballast: one per post at 14.75 kg	☐	☐
☐	buy/weight_plate_1p25	Cast iron standard plate, 1.25 lb, 1 in hole (101.6 x 10.2 mm) weight_plate_1p25	RPB1-25 seller's no. RPB1-25 Body-Solid	2	$3.33	$6.66	1 alternate · 1 drop-in DROP-IN Amazon (CAP Barbell) ASIN B002BRAN66 · $18.97 checked 2026-10-27	Uses and sourcing notes rig ×2 ballast: the 16 kg sweep	☐	☐
☐	buy/nucleo_g474re	ST Nucleo-64 NUCLEO-G474RE (the balance and power MCU: 3 FDCAN) nucleo_g474re	NUCLEO-G474RE ST eStore In stock (1-2 per order) (2026-10-27)	1	$19.90	$19.90	1 alternate · 1 drop-in DROP-IN SparkFun product page · $31.50 · In stock; Teensy 4.1 (3 CAN) as the backup MCU checked 2026-10-27	Uses and sourcing notes p0 control ×1 the balance and power MCU (three FDCAN: buses A, B, C)	☐	☐
☐	buy/xt60h_f	Amass XT60H female, 5-pack (the robot's live ends: the charge port and the 48 V branch leads; D58) xt60h_f	XT60H-F Pyrodrone available (2026-10-27)	6 buy 2 packs of 5	$0.798 $3.99 per 5	$7.98 used $4.79	1 alternate · 1 drop-in DROP-IN RaceDayQuads product page · $0.99 · available (single) checked 2026-10-27	Uses and sourcing notes p0 wiring ×6 sockets on the robot, no live pins (D58, hazard 11): the six bus branch leads' ends (W10-W15)	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/xt60h_m	Amass XT60H male, 5-pack (the devices' end: the VESC leads; D58) xt60h_m	XT60H-M 5-pack Pyrodrone available (2026-10-27)	2 buy 1 pack of 5	$0.798 $3.99 per 5	$3.99 used $1.60	1 alternate · 1 drop-in DROP-IN RaceDayQuads product page · $0.99 · available (single) checked 2026-10-27	Uses and sourcing notes p0 wiring ×2 the two VESC leads' plugs (the devices carry the pins; the RS-04 pigtails' XT60H-M come with the leg's pigtails, docs/body-next-plan.md 5.1 item 5)	☐	☐
☐	buy/xt30_f	Amass XT30U female, 8-pack (the robot's live ends: the 24 V and 12 V branch leads; D58) xt30_f	XT30U-F 8-pack Pyrodrone available (2026-10-27)	3 buy 1 pack of 8	$0.3738 $2.99 per 8	$2.99 used $1.12	1 alternate · 1 drop-in DROP-IN RaceDayQuads product page · $0.99 · available (single) checked 2026-10-27	Uses and sourcing notes p0 wiring ×3 sockets on the robot: the J501 (W21), the lidar box (W24); one spare (D76: the turret takes its power on the DT06 plug)	☐	☐
☐	buy/xt30_m	Amass XT30U male, 8-pack (the devices' end: the J501 and lidar box adapter leads; D58) xt30_m	XT30U-M 8-pack Pyrodrone available (2026-10-27)	4 buy 1 pack of 8	$0.3738 $2.99 per 8	$2.99 used $1.50	1 alternate · 1 drop-in DROP-IN RaceDayQuads product page · $0.99 · available (single) checked 2026-10-27	Uses and sourcing notes p0 wiring ×4 the devices' pins: the J501's DC-IN and the lidar box's adapter leads, two spare	☐	☐
☐	buy/radio_stop_rx	RadioMaster ER3C-i ExpressLRS 2.4 GHz PWM receiver (the radio stop, D37: one PWM channel drives the RC relay switch) radio_stop_rx	ER3C-i 2.4 GHz ELRS PWM RadioMaster available (store JSON) (2026-10-27)	1	$16.99	$16.99	1 alternate · 1 drop-in DROP-IN RadioMaster product page · $18.99 · available; RP1 V2 nano, PWM on its UART pads (ELRS 3 setting) checked 2026-10-27	Uses and sourcing notes p0 control ×1 the radio stop's receiver (D37), on the 5 V rail	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/fuse_mini58_20a	Littelfuse MINI 58V blade fuse 20 A (the RS-04 branches) fuse_mini58_20a	0997020.WXN seller's no. C207028 LCSC 1,609 (2026-10-27)	9	$0.610	$5.49	2 alternates · 2 drop-in DROP-IN element14 product page · listed (snippet) checked 2026-10-27 DROP-IN element14 product page · listed (snippet) checked 2026-10-27	Uses and sourcing notes stand ×1 bench p0 power skid ×8 the four RS-04 branches (14 AWG); four spares	☐	☐
☐	buy/fuse_mini58_10a	Littelfuse MINI 58V blade fuse 10 A (DDR-120C-24 input, the charge port) fuse_mini58_10a	0997010.WXN seller's no. C207026 LCSC 3,724 (2026-10-27)	6	$0.410	$2.46	2 alternates · 2 drop-in DROP-IN LCSC C207027 · $0.66 · 2,950 (15 A) checked 2026-10-27 DROP-IN LCSC C207027 · $0.66 · 2,950 (15 A) checked 2026-10-27	Uses and sourcing notes turret ×2 el_fuse p0 power skid ×4 the TRACO input (W16) and the J501 (W21); two spares	☐	☐
☐	buy/fuse_mini58_7a5	Littelfuse MINI 58V blade fuse 7.5 A (the turret's feed W22 and the 12 V rail W43, D66) fuse_mini58_7a5	099707.5WXN seller's no. C2680610 LCSC ships from CN 3,810 (2026-10-27)	4	$0.5766	$2.31	1 alternate · 1 drop-in DROP-IN LCSC (other suppliers) C2680610 · $0.31 · 3,000 (5-7 business days) checked 2026-10-27	Uses and sourcing notes p0 power skid ×4 the turret's branch (W22: the stop relay's contact carries 8 A) and the 12 V loads (W43); two spares	☐	☐
☐	buy/fuse_mini58_30a	Littelfuse MINI 58V blade fuse 30 A (the VESC branches) fuse_mini58_30a	0997030.WXN seller's no. C207030 LCSC ships from CN 2,894 (2026-10-27)	4	$0.440	$1.76	1 alternate · 1 drop-in DROP-IN LCSC C207029 · $0.49 · 1,743 (25 A) checked 2026-10-27	Uses and sourcing notes p0 power skid ×4 the two VESC branches (12 AWG; the VESCs' battery current limit 20 A); two spares	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/gh_header_2p	JST GH 1.25 mm 2-pin header, top entry (SMT) gh_header_2p	JST BM02B-GHS-TBT(LF)(SN) (LCSC C161690) seller's no. JST BM02B-GHS-TBT(LF)(SN) (LCSC C161690) LCSC ships from CN 70,630 in stock (2026-10-27)	4	$0.0897	$0.36	1 alternate · 1 drop-in DROP-IN HQonline BM02B-GHS-TBT(LF)(SN) · $0.10 checked 2026-10-27	Uses and sourcing notes electronics ×4 the CAN junction: mates each RS-04's GH1.25-2P plug (two used, two spare)	☐	☐
☐	buy/screw_m8x16_bh	M8 x 16 button head socket cap screw screw_m8x16_bh	13-8316 seller's no. 13-8316 80/20 Inc.	16	$0.610	$9.76	1 alternate · 1 drop-in DROP-IN 80/20 Inc. 75-3422 · $1.14 checked 2026-10-27	Uses and sourcing notes rig ×16 with the extra M8 drop-in nuts	☐	☐
☐	buy/screw_m6x40	Black-Oxide Alloy Steel Socket Head Screw, US Origin, M6 x 1 mm Thread Size, 40 mm Long screw_m6x40	6544 seller's no. 6544 Bolt Depot no count shown; orderable (Add button, not marked 'Temporarily unavailable') (2026-10-27)	8	$0.260	$2.08	1 alternate · 1 drop-in DROP-IN Fastenal 1139571 · $21.50 / 50 · Local DC Inventory; Direct Ship: Ships TOMORROW checked 2026-10-27	Uses and sourcing notes bench ×8 the dyno: the two lug plates through the 19 mm board	☐	☐
☐	buy/bolt_m12x80	Hex bolt M12 x 80, fully threaded (DIN 933) bolt_m12x80	6132 seller's no. 6132 Bolt Depot no count shown; orderable (Add button, not marked 'Temporarily unavailable') (2026-10-27)	1	$1.99	$1.99	1 alternate · 1 drop-in DROP-IN Fastenal 925206494 · $15.10 / 10 · FNL Supply Chain (not in the local DC); Direct Ship: Ships TOMORROW checked 2026-10-27	Uses and sourcing notes bench ×1 the press cage's press bolt (B3)	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/screw_m12x30	Alloy steel socket head screw M12 x 30, class 12.9 screw_m12x30	6604 seller's no. 6604 Bolt Depot no count shown; orderable (Add button, not marked 'Temporarily unavailable') (2026-10-27)	2	$0.960	$1.92	1 alternate · 1 drop-in DROP-IN Fastenal 1139654 · $38.00 / 25 · Local DC Inventory; Direct Ship: Ships TOMORROW checked 2026-10-27	Uses and sourcing notes bench ×2 the press cage: the load cell to the top beam; the anvil on the cell (B3)	☐	☐
☐	buy/nut_m6_nylock	Nylon-insert lock nut M6 nut_m6_nylock	4804 seller's no. 4804 Bolt Depot no count shown; orderable (Add button, not marked 'Temporarily unavailable') (2026-10-27)	8	$0.100	$0.80	1 alternate · 1 drop-in DROP-IN Fastenal 1140155 · $9.00 / 100 · Local DC Inventory; Direct Ship: Ships TOMORROW checked 2026-10-27	Uses and sourcing notes bench ×8 the dyno: under the board	☐	☐
☐	buy/screw_m8x40	Alloy steel socket head screw M8 x 40, class 12.9 screw_m8x40	6565 seller's no. 6565 Bolt Depot no count shown; orderable (Add button, not marked 'Temporarily unavailable') (2026-10-27)	2	$0.380	$0.76	1 alternate · 1 drop-in DROP-IN Fastenal 1139591 · $35.50 / 50 · Local DC Inventory; Direct Ship: Ships TOMORROW checked 2026-10-27	Uses and sourcing notes bench ×2 B4: the M8 insert under test	☐	☐
☐	buy/nut_m12	Hex nut M12, DIN 934 nut_m12	4790 seller's no. 4790 Bolt Depot no count shown; orderable (Add button, not marked 'Temporarily unavailable') (2026-10-27)	2	$0.240	$0.48	1 alternate · 1 drop-in DROP-IN Fastenal 1140309 · $8.25 / 25 · Local DC Inventory; Direct Ship: Ships TOMORROW checked 2026-10-27	Uses and sourcing notes bench ×2 in the press base (the press bolt) and in the puller (the stud)	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/washer_m12	Flat washer M12, DIN 125 washer_m12	4531 seller's no. 4531 Bolt Depot no count shown; orderable (Add button, not marked 'Temporarily unavailable') (2026-10-27)	2	$0.080	$0.16	1 alternate · 1 drop-in DROP-IN Fastenal 1140359 · $4.00 / 25 · Local DC Inventory; Direct Ship: Ships TOMORROW checked 2026-10-27	Uses and sourcing notes bench ×2 under the slider (on the press bolt's tip); a spare	☐	☐
☐	buy/usb-can-canable2	UCAN Board Based on STM32F072 USB to CAN Adapter Support with CAnable / CandleLight / Klipper firmware usb-can-canable2	UCAN (SKU MM-U2CAN-0001) seller's no. MM-U2CAN-0001 (variant 42187830067375) FYSETC available: true (Shopify; no count exposed) (2026-10-27)	1	$8.00	$8.00	See note below	Uses and sourcing notes stand ×1 bench	☐	☐
Alternates: 5 alternates · 3 drop-in DROP-IN BIQU / BIGTREETECH 1030000107 (variant 39762747949154, '1 Pc / U2C V2.1 -3 CAN Output') · $23.99 · available: true (Shopify; no count exposed) checked 2026-10-27 DROP-IN Amazon ASIN B0956NV6CM (UPC 781520762707) · $29.99 · Only 15 left in stock - order soon checked 2026-10-27 DROP-IN Openlight Labs CANable 2.0 (SKU PF-CANABLE-2.0) · $35.00 · Out of stock checked 2026-10-27 CHANGES BIQU Equipment BIGTREETECH U2C V2.1 (variant '1 Pc / U2C V2.1 -3 CAN Output', SKU 1030000107) · $23.99 · Orderable Board size not checked against the stand's 50 x 20 x 12 mm envelope, and it has several CAN outputs ('-3 CAN Output'). The listing is shared with the EBB boards, so pick the U2C variant. Firmware is BTT's candleLight fork (gs_usb), so Linux gets can0 without slcand. checked 2026-10-27 CHANGES Grid Connect PEAK-System PCAN-USB IPEH-002021 (Grid Connect GC-CAN-USB); opto-isolated IPEH-002022 (GC-CAN-USB-ISO) · $256.00 (no isolation); $314.00 (with isolation) · Orderable An enclosed dongle on a cable, not a board on the stand floor, so the back-wall USB-C slot and the 50 x 20 x 12 mm envelope do not apply. From general knowledge, not the page: it has a D-Sub 9 CAN connector, so it needs a DB9-to-wire lead, and it uses Linux's peak_usb SocketCAN driver instead of gs_usb or slcan. About 30 times the price of the UCAN. checked 2026-10-27									
☐	buy/jack_3p5	3.5 mm panel jack jack_3p5	Switchcraft 35PM2A seller's no. Switchcraft 35PM2A Master Electronics	1	$6.95	$6.95	2 alternates · 2 drop-in DROP-IN Markertek Calrad 30-493A checked 2026-10-27 DROP-IN Canford Canford 43-259 miniature jack socket, 3.5 mm 3-pole checked 2026-10-27	Uses and sourcing notes electronics ×1 the pendant's plug on the pod's panel	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/gh_lead_2p_300	JST GH 1.25 pre-crimped lead, socket to socket, 300 mm (CAN, UART) gh_lead_2p_300	AGHGH28K305 TLC Electronics 12,900 (2026-10-27)	8	$0.520	$4.16	1 alternate · 1 drop-in DROP-IN TLC Electronics product page · $0.48 · 6,000 (150 mm) checked 2026-10-27	Uses and sourcing notes p0 wiring ×8 CAN and I2C leads to the transceivers and the INA228	☐	☐
☐	buy/amass-xt60e-m	XT60E-M panel-mount plug (male), 2 screw holes amass-xt60e-m	Amass XT60E-M seller's no. Amass XT60E-M Maplin ships from UK	2	$1.50	$3.00	2 alternates · 2 drop-in DROP-IN LCSC XT60E-M.G.Y (C3040679) · $0.61 · In stock (3,091) checked 2026-10-27 DROP-IN PyroDrone F3-13155-BLK · $11.99 / 10 · in stock checked 2026-10-27	Uses and sourcing notes electronics ×1 the pod's inlet: the pack lead carries the female stand ×1 bench	☐	☐
☐	buy/fuse_mini58_5a	Littelfuse MINI 58V blade fuse 5 A (DDR-60L-12 input; the 24 V J501 and turret branches) fuse_mini58_5a	0997005.WXN Farnell 5,995 in stock, 10-14 business days from US stock (search result, 2026-10-27) (2026-10-27)	2	$0.980	$1.96	1 alternate · 1 drop-in DROP-IN Dasenic product page · $0.39 · 17,902 in stock, $0.3878 at 25+ (search result, 2026-10-27) checked 2026-10-27	Uses and sourcing notes p0 power skid ×2 the INA228's VBUS sense lead (W32, 22 AWG); one spare	☐	☐
☐	material/pa6-cf	Bambu Lab PA6-CF pa6-cf	pa6-cf Bambu Lab	6.07 kg buy 7 kg	$79.99 / kg	$559.93 used $485.57	Use the material specified by the print plan.	Uses and sourcing notes 6.07 kg in the included print plans: leg 3.08 kg · rig 1.87 kg · turret 0.56 kg · body 0.57 kg	☐	☐
☐	material/petg	Bambu Lab PETG Basic petg	petg Bambu Lab	3.19 kg buy 4 kg	$13.99 / kg	$55.96 used $44.57	Use the material specified by the print plan.	Uses and sourcing notes 3.19 kg in the included print plans: rig 2.39 kg · turret 0.59 kg · body 0.21 kg	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	material/tpu95	Bambu Lab TPU 95A HF tpu95	tpu95 Bambu Lab	0.45 kg buy 1 kg	$41.99 / kg	$41.99 used $18.79	Use the material specified by the print plan.	Uses and sourcing notes 0.45 kg in the included print plans: leg 0.06 kg · rig 0.37 kg · turret 0.01 kg · body 0.01 kg	☐	☐
☐	material/tpu95-gyroid	Bambu Lab TPU 95A HF, printed as a gyroid crush lattice (about 15 % infill, 2 walls, 1 mm skins) tpu95-gyroid	tpu95-gyroid Bambu Lab	0.26 kg buy 1 kg	$41.99 / kg	$41.99 used $10.75	Use the material specified by the print plan.	Uses and sourcing notes 0.26 kg in the included print plans: body 0.26 kg	☐	☐
☐	buy/?base_board_inserts	base_board_inserts ? base_board_inserts	NO PART NUMBER	1	NO PRICE	—	NO ALTERNATE	Uses and sourcing notes turret ×1 base_board_inserts: drawn in cad/turret.py, not on the bench list yet NO CATALOGUE KEY	☐	☐
☐	buy/?base_board_screws	base_board_screws ? base_board_screws	NO PART NUMBER	1	NO PRICE	—	NO ALTERNATE	Uses and sourcing notes turret ×1 base_board_screws: drawn in cad/turret.py, not on the bench list yet NO CATALOGUE KEY	☐	☐
☐	buy/?base_carrier_inserts	base_carrier_inserts ? base_carrier_inserts	NO PART NUMBER	1	NO PRICE	—	NO ALTERNATE	Uses and sourcing notes turret ×1 base_carrier_inserts: drawn in cad/turret.py, not on the bench list yet NO CATALOGUE KEY	☐	☐

Done	Part ID	Part	Supplier part number	Quantity needed	Unit price	Order cost	Alternates	Part details	Ordered	In hand
☐	buy/?base_carrier_screws	base_carrier_screws ?base_carrier_screws	NO PART NUMBER	1	NO PRICE	—	NO ALTERNATE	Uses and sourcing notes turret ×1 base_carrier_screws: drawn in cad/turret.py, not on the bench list yet NO CATALOGUE KEY	☐	☐
☐	buy/?hopper_inserts	hopper_inserts ?hopper_inserts	NO PART NUMBER	1	NO PRICE	—	NO ALTERNATE	Uses and sourcing notes turret ×1 hopper_inserts: drawn in cad/turret.py, not on the bench list yet NO CATALOGUE KEY	☐	☐
☐	buy/?laser_inserts	laser_inserts ?laser_inserts	NO PART NUMBER	1	NO PRICE	—	NO ALTERNATE	Uses and sourcing notes turret ×1 laser_inserts: drawn in cad/turret.py, not on the bench list yet NO CATALOGUE KEY	☐	☐
☐	buy/?laser_screws	laser_screws ?laser_screws	NO PART NUMBER	1	NO PRICE	—	NO ALTERNATE	Uses and sourcing notes turret ×1 laser_screws: drawn in cad/turret.py, not on the bench list yet NO CATALOGUE KEY	☐	☐